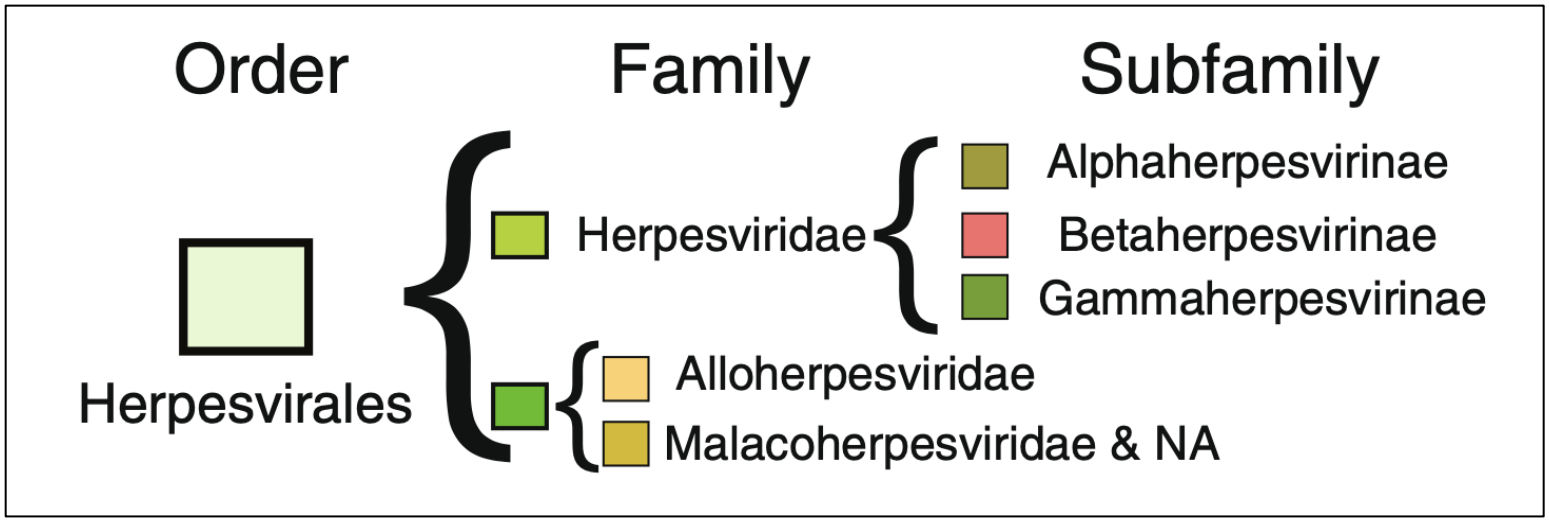
Swine Herpesvirus Diseases

Chi-Fei Kao (高啓霏)

2026.10.01

Herpesviruses



Andrade-Martínez et al., 2019

Host species:

- *Herpesviridae*: Reptiles, birds, and **mammals**
- *Alloherpesviridae*: amphibia and fish
- *Malacoherpesviridae*: mollusks (軟體動物)

Characteristics of Herpesviridae

- Similar virion morphology (from the inside out):
  - ✓ Condensed DNA core,
  - ✓ Icosahedral capsid,
  - ✓ Proteinaceous tegument,
  - ✓ Lipid envelope containing **glycoproteins**.

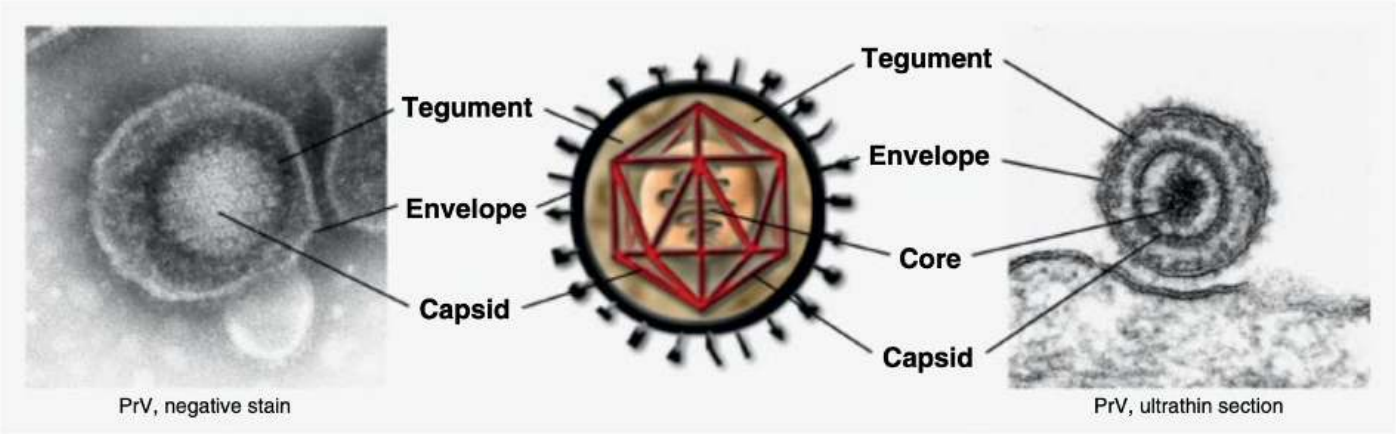
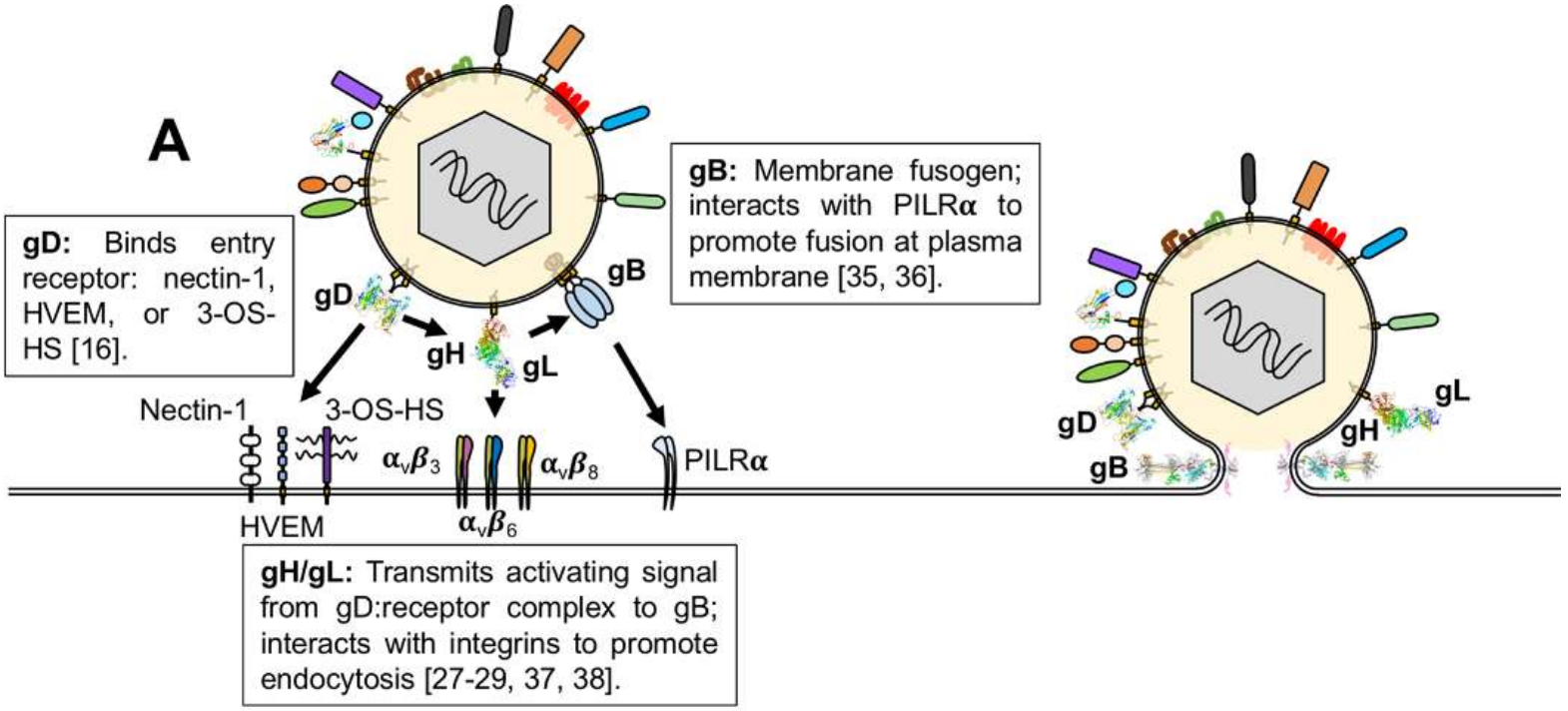
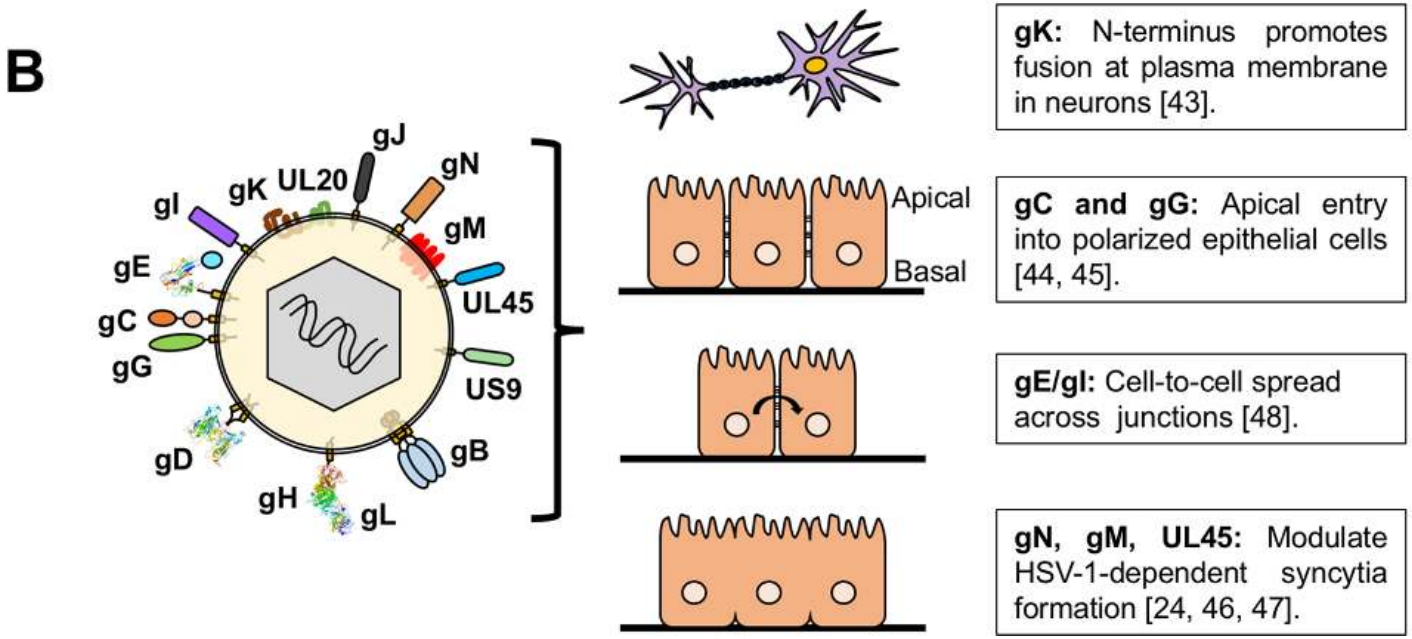


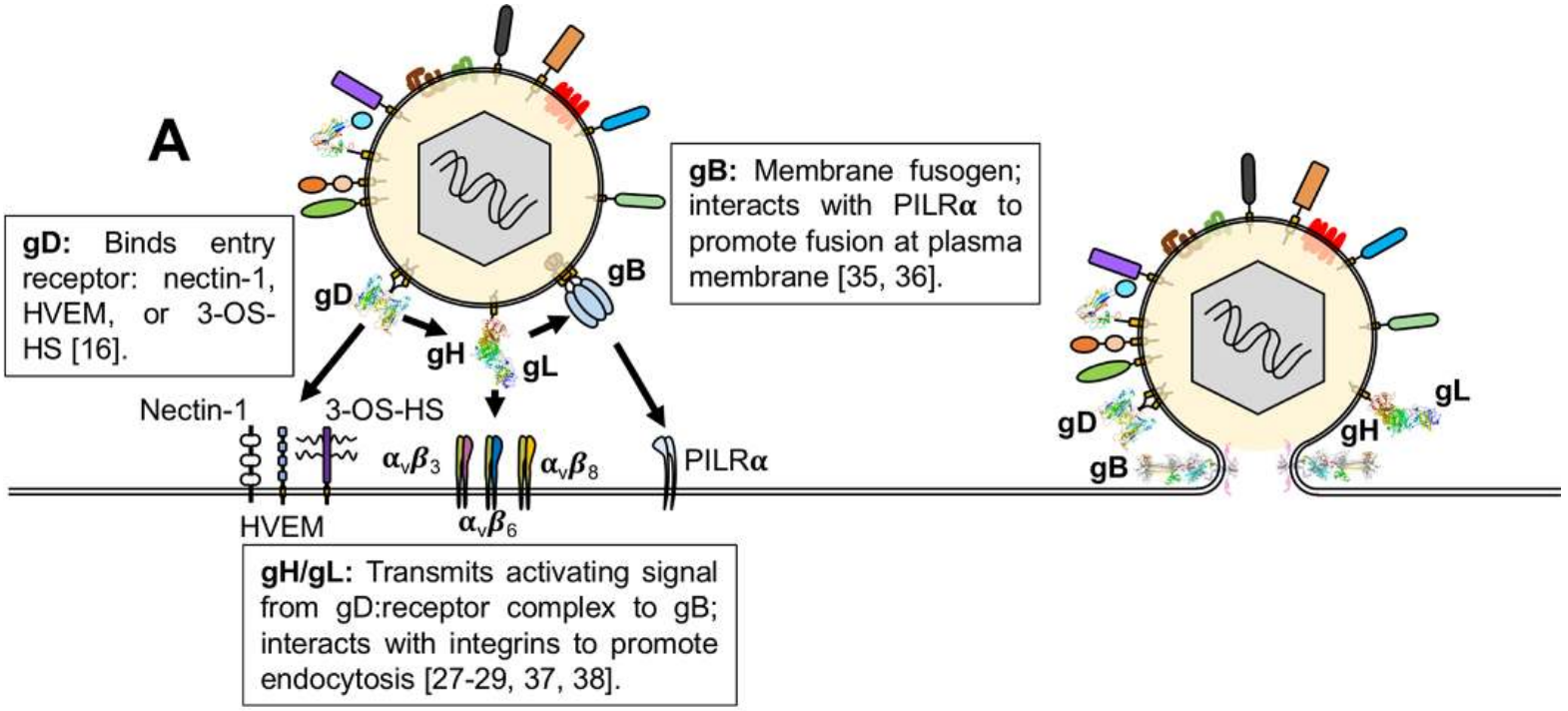
Figure 35.1 Morphology of a typical herpes virion (PRV). *Source:* Mettenleiter et al. (2008). Reproduced with permission of Elsevier.

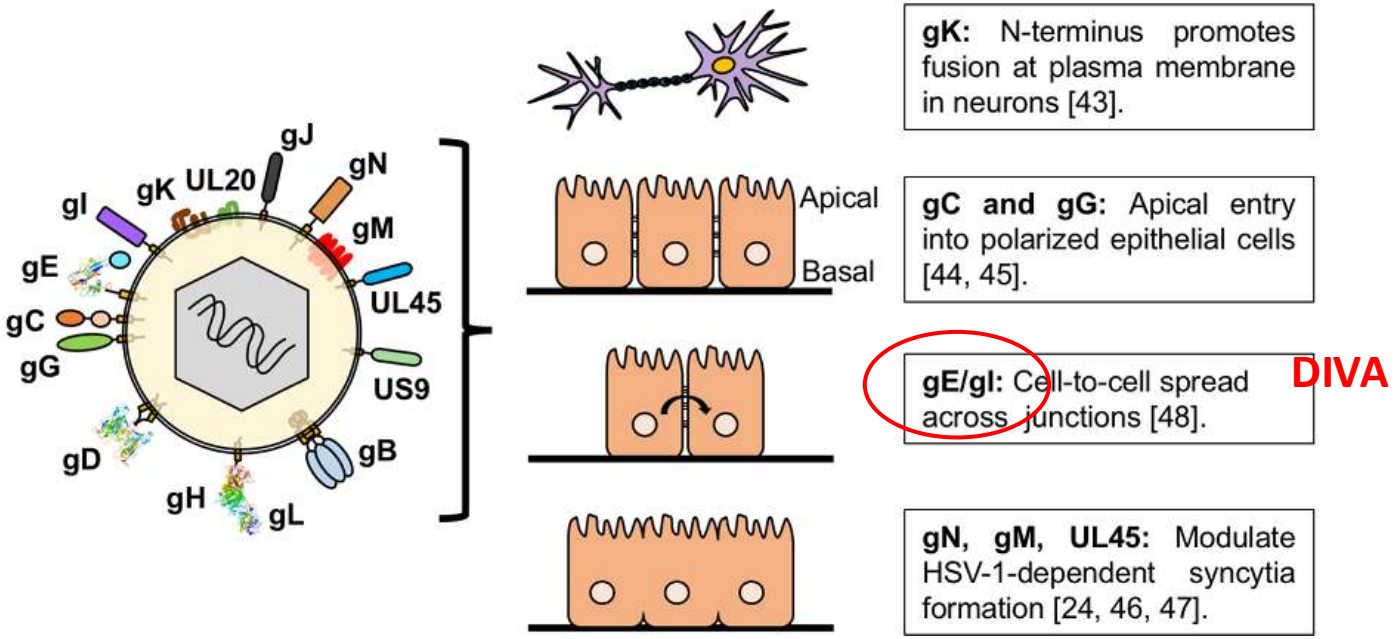
Diseases of Swine, 11th Edition





Hilterbrand & Heldwein, 2019





Hilterbrand & Heldwein, 2019

Characteristics of Herpesviridae

- Infection has **lytic** and **latent phases**
- In the latency, the viral gene expression is restricted and no viral progeny is produced → lifelong association with the host.
- **Replication** (lytic infection):
  - ✓ Kinetic transcription: immediate-early, early & late
  - ✓ Rolling-circle replication mechanism of viral DNA synthesis in the nucleus
  - ✓ Budding of encapsidated viral DNA through penetration of nuclear membrane
  - ✓ Assembly of tegument proteins and secondary envelopment in a Golgi or post-Golgi compartment.
  - ✓ Release from cell by exocytosis

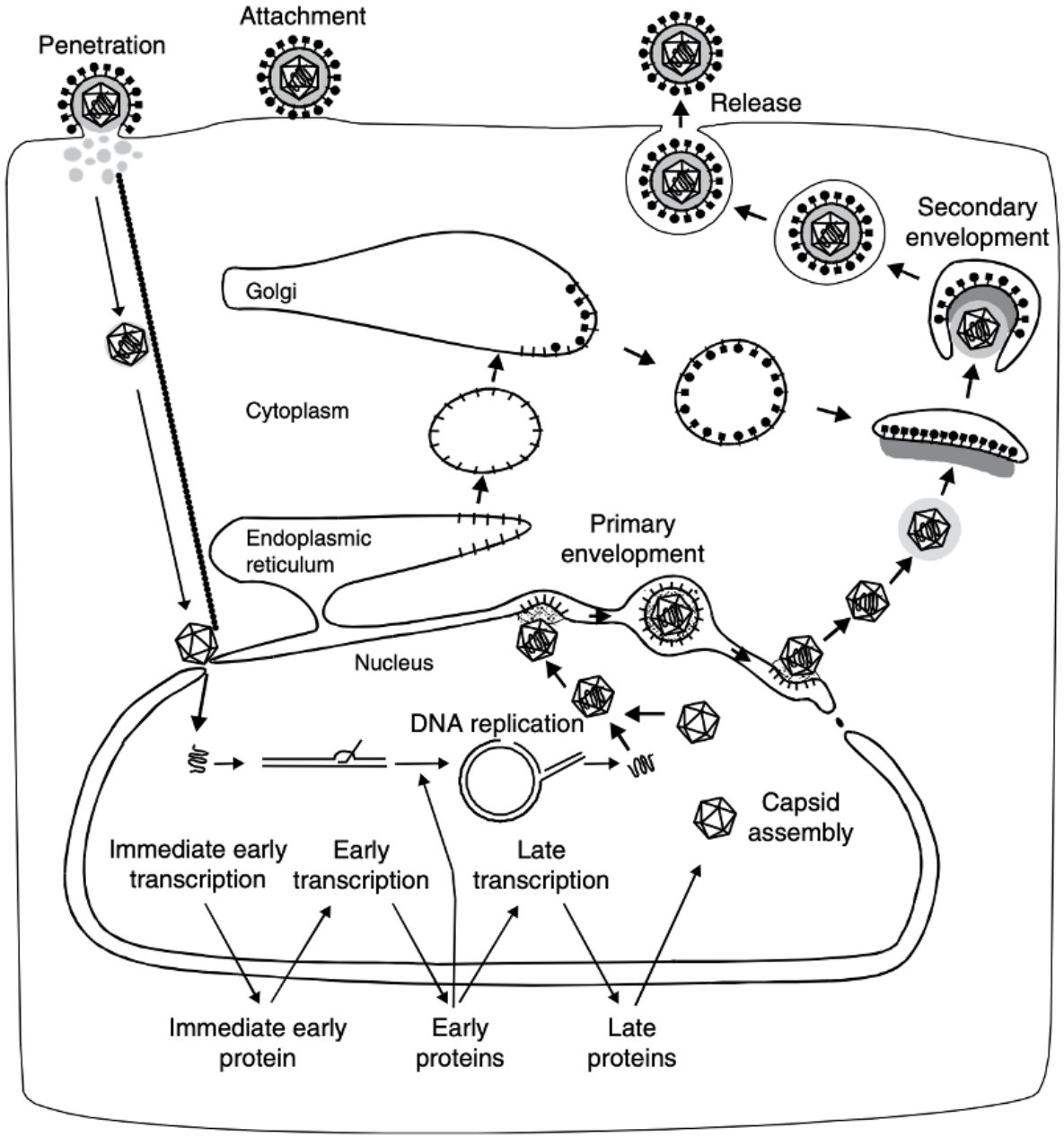


Figure 35.2 Herpesvirus replication cycle. *Source:* Mettenleiter (2004). Reproduced with permission of Elsevier.

Table 35.1 Herpesviruses of swine.[a]

New taxonomic name	Trivial name	Genus	Subfamily
Suid alphaherpesvirus 1	Pseudorabies virus (Aujeszky's disease virus)	*Varicellovirus*	*Alphaherpesvirinae*
Suid betaherpesvirus 2	Porcine cytomegalovirus	Unassigned	*Betaherpesvirinae*
Suid gammaherpesvirus 3	Porcine lymphotropic herpesvirus 1	*Macavirus*	*Gammaherpesvirinae*
Suid gammaherpesvirus 4	Porcine lymphotropic herpesvirus 2	*Macavirus*	*Gammaherpesvirinae*
Suid gammaherpesvirus 5	Porcine lymphotropic herpesvirus 3	*Macavirus*	*Gammaherpesvirinae*
Ovine gammaherpesvirus 2	Sheep-associated malignant catarrhal fever virus	*Macavirus*	*Gammaherpesvirinae*

[a] Whereas pigs are main/sole hosts for suid alphaherpesvirus 1, suid betaherpesvirus 2, and suid gammaherpesviruses 3-5, OvHV-2 is primarily found in sheep but causes spillover infections in pigs.

Diseases of Swine, 11th Edition

Swine herpesviruses are present in all three subfamilies.

Pseudorabies (PR)

遭狗咬傷　台東婦人染疱疹病毒引起的「假性狂犬病」

免費休閒小遊戲，免下載免安裝，隨點即玩!

0
讚

我們想讓你知道…

「假性狂犬病」不會傳染人也不會發病，不要過度恐慌



▲真正控制犬隻狂犬病的有效方法是讓犬隻接種狂犬病疫苗。(圖／動保處提供)

地方中心／台東報導

台東縣卑南鄉一名黃姓婦人，7月25日被養在菜園的黑狗咬傷，小狗檢體並未驗出狂犬病毒，農委會發現，婦人染上的是屬於**疱疹病毒的「假性狂犬病」，好發於豬隻，不會傳染人也不會發病**，研判可能是婦人餵狗吃生豬肉因此感染，但遺憾的是，在這之前，婦人已同意讓狗安樂死。

https://www.ettoday.net/news/20130807/252134.htm

Introduction

- A swine viral disease with manifestation in **nerve**, **respiratory** and **reproductive** system.
- The first clinical PRV was reported in 1813 in cattle.
- Also known as Aujeszky's disease, named after the Hungarian veterinarian Dr. Aladár Aujeszky who linked the disease in cattle, dogs, and cats in 1902. 奥耶兹基氏病
- **Wide host range** while **pig is the only natural host**.

Introduction

Significance in swine industry:

- Devastating losses by **fatal infection of piglets** and **abortions** in pregnant animals.
- Infection of fetuses results in resorption, mummification, or abortion.
- **Vaccination (DIVA)** provided the basis for cost-effective control and, eventually, PRV eradication.

Etiology

Pseudorabies virus (PRV) or suid herpesvirus 1 (SuHV–1)

- Genus *Varicellovirus*, subfamily of *Alphaherpesviridae*
- **Enveloped** & approximately 200 nm in diameter
- Double-stranded sense DNA genome, approximately 143 kb encoding at least 72 genes
- The genome consists of two unique regions (UL and US) and two repeat sequences (IRS and TRS).
- Only **one serotype**

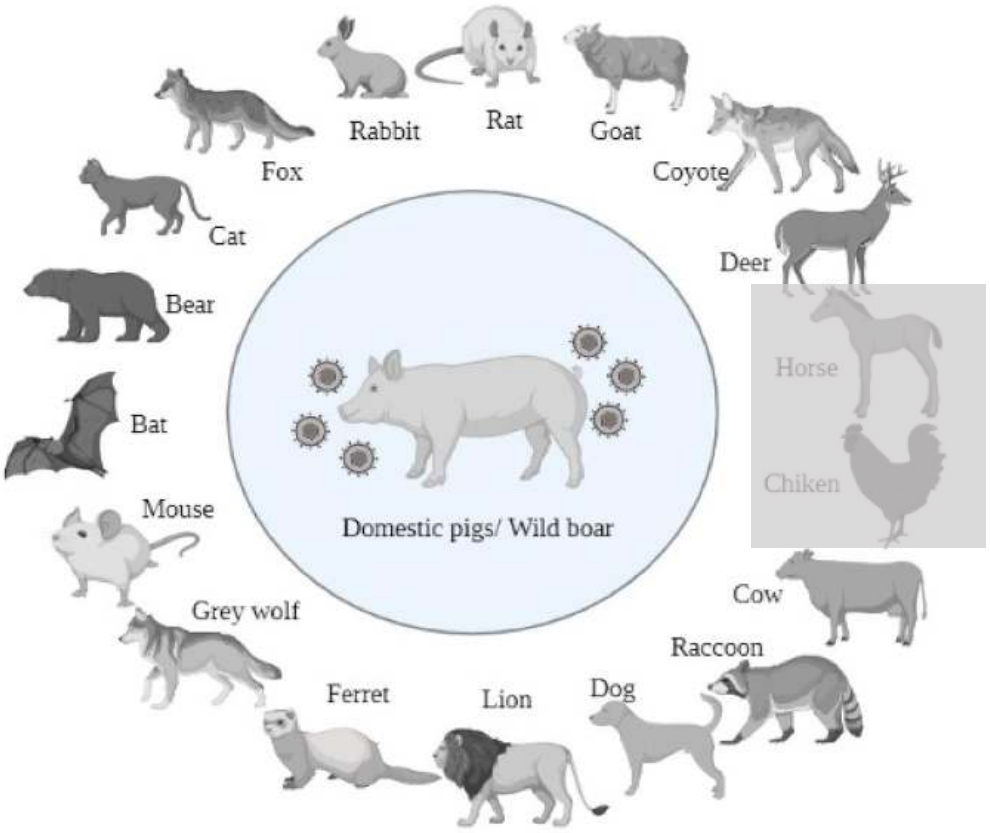
Epidemiology

- Once worldwide distributed but **eradication has been achieved in industry** in many regions by efficacious **DIVA vaccination**.
- PRV in wild boars represent a constant danger for reintroduction of PRV into free herds and regions.
- Vaccination is prohibited in PRV-free countries.
- Notably, the **emerging PRV variants** in China since 2011 escape immunoprotection induced by the traditional live-attenuated Bartha-K61 vaccines.

Epidemiology

Susceptible species:

- Only pigs survive the productive infection → considered the only natural host for PRV
- PRV can **naturally infect** cattle, sheep, cats, dogs, rabbit, mice, and rats, causing **fatal neurological disease**.
- The “dead-end” hosts shed PRV for only a short time → **limited lateral transmission**



Bo et al., 2022

Epidemiology

Susceptible species:

- **Dogs** appear to be highly susceptible. Infection with PRV usually resulted from consuming raw meat or in close contact with swine (eg hunting dogs).
- **Rabbits** also show high susceptibility and is widely used as experimental animals for **PRV isolation test**.

Ferrara *et al. BMC Veterinary Research* (2024) 20:323
https://doi.org/10.1186/s12917-024-04189-3

BMC Veterinary Research

CASE REPORT Open Access

A pseudorabies outbreak in hunting dogs in Campania region (Italy): a case presentation and epidemiological survey

Gianmarco Ferrara[1*], Ugo Pagnini[1], Antonio Parisi[1], Maria Grazia Amoroso[2], Giovanna Fusco[2], Giuseppe Iovane[1] and Serena Montagnaro[1]



Epidemiology

Resistant species:

- In general, **avian**, **horses** and **higher primates** are resistant to pseudorabies virus infection.
- There are (a few) reports of PRV infection in horses or chickens under either experimental or natural conditions.

Zoonotic Potential?

Year	Numbers	Symptoms	Contacted Animals
1914	2	Swelling, reddening, intense itching	Cat
1940	2	Pruritis, erythema, and pain around the wound	Dog
1963	2	Throat pain, weakness in the legs,	Dog
1983	1	Pain in togue, hypersalivation, dysphagia, headache, arthralgia	Cat
1986	2	Tension in the mouth, nose, and throat; perception of strange smells and taste	Cat, or other domestic animals
1992	6	Pruritus in the palms, lower and upper arms, shoulders, and back	Cow
2017	1	Endophthalmitis, fever, headaches	Pig
2018	4	Encephalitis	Pig
2019	5	Encephalitis	Pig
2019	1	Encephalitis	Pig
2019	1	Encephalitis, fever, headache, seizure	unknown
2019	1	Encephalitis	Pig
2020	1	Encephalitis	Pig
2020	6	panencephalitis	Pig
2021	1	Retinitis	Pig
2021	1	Encephalitis, retinal vasculitis, fever	Pig
2021	1	Encephalitis, Endophthalmitis	Pig
2022	2	Encephalitis, seizures, endophthalmitis	Pig

Bo et al., 2022

Zoonotic Potential?

- **Inconclusive, but…**

  ✓Increasing numbers of human unexplained encephalitis and endophthalmitis associated with **PRV-DNA positivity** are reported. **Is genome-positive enough?*

  ✓Supportive *in vivo* and *in vitro* data were published:

  -- A PRV variant-like strain, hSD-1/2019, was **isolated** from CSF of a vet showing neurological symptoms and contact history with diseased pigs.

  -- The human-associated PRV variants were shown to be able to adapt to human cell lines *in vitro* and displays increased intercellular/syncytial spread in these cells.

Spillover of pseudorabies virus variants to humans: an urgent call for pseudorabies eradication in domestic pigs

Tong-Yan Wang, Chao Li, Xue-Hui Cai*, Yongjing Shan*, Yan-Dong Tang*

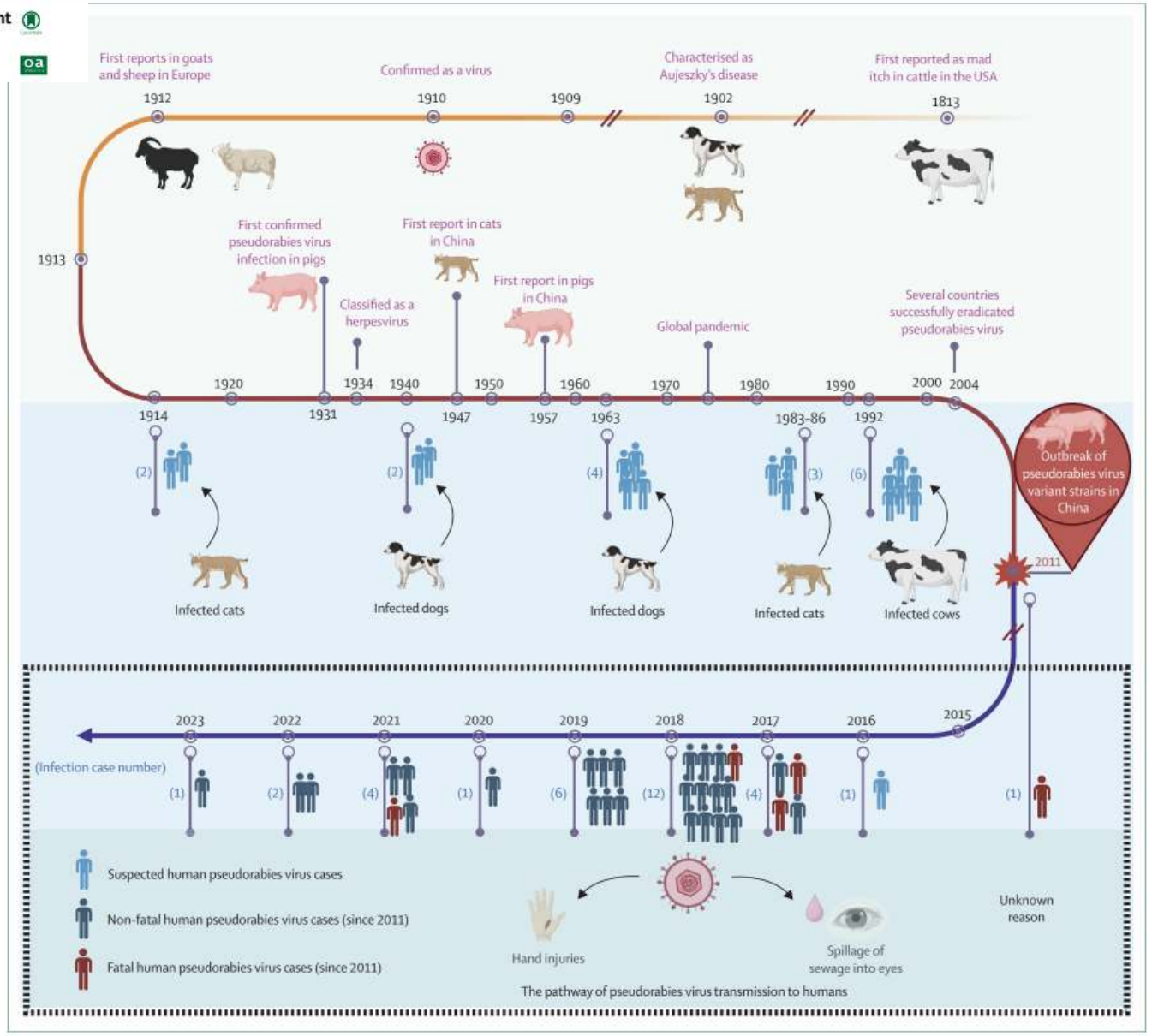


Figure 1: **Historical timeline of pseudorabies discovery and key milestones**
The timeline shows the major events in the identification and characterisation of pseudorabies virus, from the initial description of mad itch in cattle in 1813 to the confirmation of its viral cause and classification as a herpesvirus. Key milestones include Aladár Aujeszky's differentiation from rabies in 1902 and the outbreak of pseudorabies virus variant strains in swine in 2011. The lower section shows suspected and confirmed human infections with pseudorabies virus.

Transmission

PRV are present in **all body secretions, excretions, and aerosols.**

- **Direct contact**
  - ✓**Oral, nasal** or conjunctival mucosa are the main entry routes.
- **Indirect contact** with contaminated fomites
  - ✓Carcasses, meat, water, beddings…
- **Aerosols**
- **Venereal route**
  - ✓Dominant within feral swine and wild boars.

野放家豬 野豬

- **Transplacental** mainly in the last gestation
- **Ingestion** of colostrum



Nose-to-nose contact

Transmission

- Rather high quantities of virus, for example, $>1\times10^4–10^5$ $TCID_{50}$, are necessary to infect animals, except piglets (1×10^2 $TCID_{50}$)
  → **not very contagious.**
- After recovery, **pigs remain carriers** of latent virus to be reactivated after stress or hormone-stimulation.

Viral Stability

- PRV is **rather resistant** to environmental conditions.
- **Stable at pH 4–12**; higher acidic or alkaline conditions take 2–4 h to inactivate PRV .
- Relatively heat resistant and stable at normal or low temperatures.

Evolved from experimental condition

- Strain-dependent temperature-sensitivity of PRV occurs.
- Maturation of pig meat at 4 °C does not inactivate the virus. However, in meat, the virus is believed to be inactivated at −18 °C within 40 days and after heat treatment of meat and meat products at 80°C.

Pathogenesis

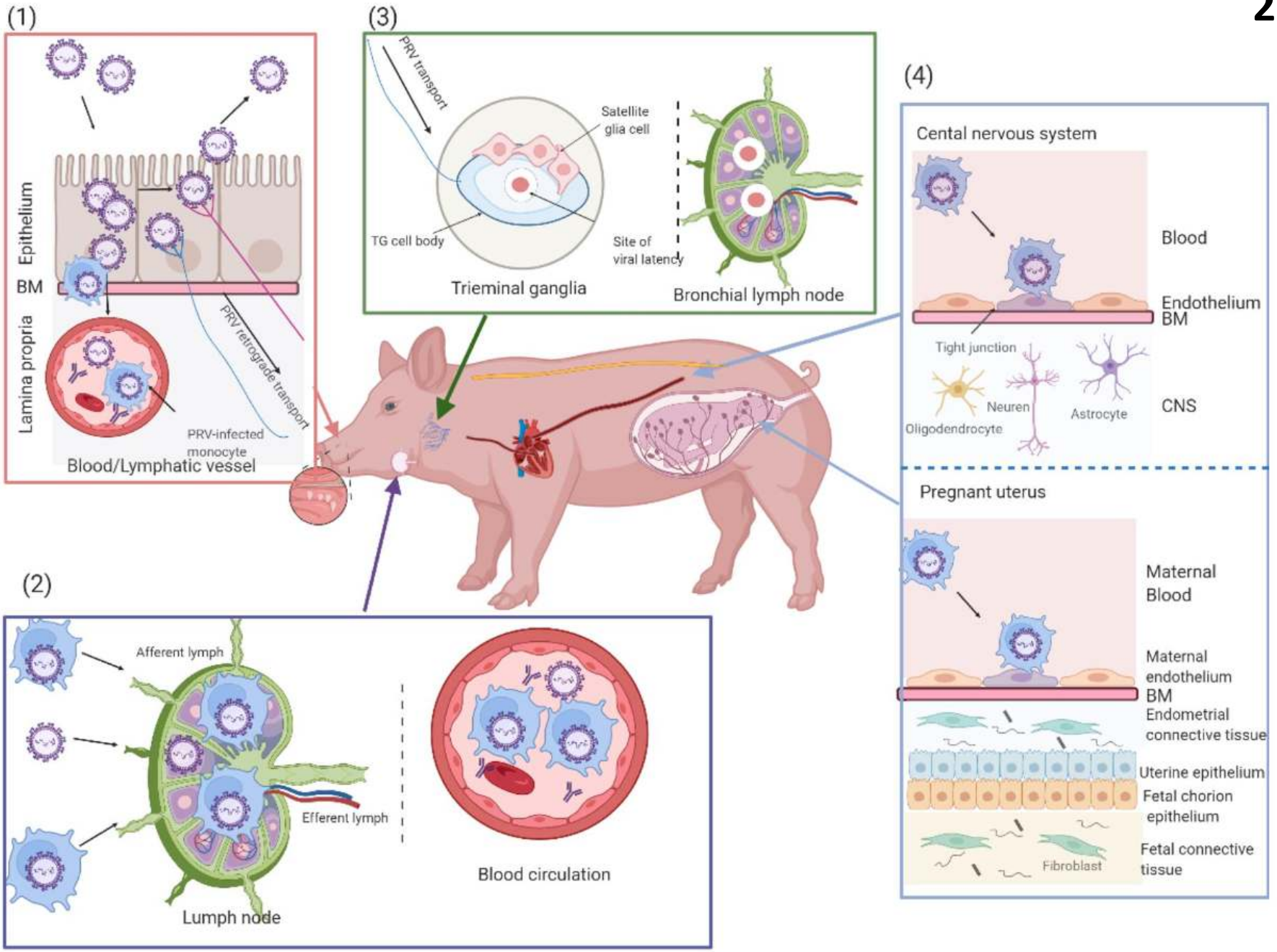
- PRV replicate primarily in **epithelial cells of the upper respiratory tract** → tonsil and local lymph nodes →

    1. Cell bodies of sensory neurons by **axonal retrograde transport**, in particular the olfactory, trigeminal, and glossopharyngeal nerves → then CNS
    舌咽神經
    2. Other organs by **secondary cell-free or cell-associated viremia mainly in monocytes,** and then replicates in epithelia, vascular endothelium, lymphocytes, and macrophages.

Wide tissue tropism

Pathogenesis

- **Trigeminal ganglia**, sacral ganglia, and tonsils are considered prime sites of latency in pigs.
- Sacral ganglia as the most common sites of PRV latency in feral swine supported the hypothesis that PRV are mainly transmitted venereally.



Zheng et al., 2022

PRV Membrane Glycoproteins

- 11 glycoproteins are expressed during replication

Table II. Properties of pseudorabies virus glycoproteins.

	Designation[a]	Gene[b]	Essential	Virion component	Attachment	Penetration	Cell-cell spread	Neuronal spread
nAb	gB (gII)	UL27	+	+	–	+[c]	+	+
nAb	gC (gIII)	UL44	–	+	[+][d]	–	–	–
nAb	gD (gp50)	US6	+	+	[+]	+	–	–
	gE (gI)	US8	–	+	–	–	[+]	[+]
	gG (gX)	US4	–	–	–	–	–	–
	gH	UL22	+	+	–	+	+	+
	gI (gp63)	US7	–	+	–	–	[+]	[+]
	gK	UL53	+	+	–	–	+	?[e]
	gL	UL1	+	+	–	+	+	?
	gM	UL10	–	+	–	–	[+]	–
	gN	UL49.5	–	+	–	[+]	–	–

(a) The old nomenclature of PrV glycoproteins has been added in parentheses.
(b) Gene designation according to the HSV-1 homologue.
(c) + indicates an essential function.
(d) [+] indicates a nonessential or modulating function.
(e) ?: no information available.

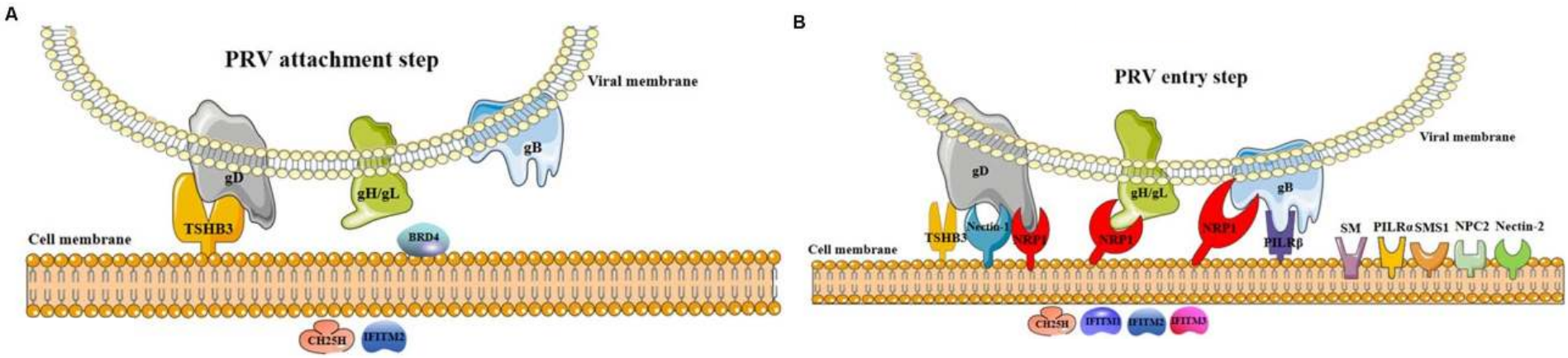
PRV Essential Glycoproteins

Binding

- **gD**: Attachment to the cellular receptor
- **gH/gL**: Transit signals from gD-binding to cellular receptor to gB to regulate virus-cell fusion
- **gB**: A class III viral fusion protein, anchoring fusion loops into the cell membrane to mediating fusion

Release

- **gK**: Prevent re-entry of the released virions



Tan et al., 2024

PRV Nonessential Glycoproteins

- **gC**: Initiate attachment to cells by binding to heparan sulfate but is dispensable for the entry process

Heterodimer

- **gE**: Promote cell fusion and is important for neuroinvasion
- **gI**: Assistant role in virion release and cell-cell spreading

- **gG**: Immune regulation by binding to host chemokines

Heterodimer

- **gN:** Dispensable for viral penetration and propagation in the nervous systems of adult mice
- **gM:** Dispensable for viral penetration and propagation in the nervous systems of adult mice

Pathogenesis

- PRV strains differ in virulence and tissue tropism widely.
  - ✓**More virulent, more neuroinvasive.**
- Virulence determinants includes
  - ✓Membrane glycoproteins: gE, gI, gC
  - ✓Virus-encoded enzymes: Thymidine kinase (TK), dUTPase, ribonucleotide reductase
  - ✓Nonessential capsid-associated proteins: UL21

→ Good targets for vaccine designation

How?

PRV Latency

- Pigs surviving a PRV infection become latently infected.
- Latency is established primarily in the neurons of the **trigeminal** and **sacral ganglia** but also in the **tonsils**.
- No infectious virus is produced during latency, but viral genomic DNA persists extrachromosomally.
- Because of the potential to reactivate and shed infectious PRV, **latently infected animals are a major threat to disease control.**
- Reactivation can occur under stress (transport, handling, temperature) or hormonal (gestation, farrowing) stimulation.

Clinical Presentation

- **Vary;** Age, immune status of the pig, the route of infection, and the virulence of the PRV strain affect the clinical severity, morbidity and mortality.
- Infection in fully susceptible swine result in high morbidity and mortality. Young & naive
- Incubation period: vary; normally ranges from 1 to 8 days, but may take up to 3 weeks.

Clinical Presentation – Pigs

- **Clinical signs:** high fever, anorexia, dyspnea, excessive salivation, trembling, incoordination, dog-sitting position, paddling, and respiratory signs (older animals).

✓ Neonatal piglets: sudden death

✓ 2-3 w nursing piglets: severe signs with **CNS** involvement

✓ 3-9 w pigs: neurological signs with age-dependent resistance respiratory signs in older pigs.

✓ >10 w pigs: **respiratory signs** predominate with high morbidity

-- Pigs may die from secondary bacterial pneumonia
-- Coinfection with other viruses may cause fatal proliferative and necrotizing pneumonia (PNP)



https://www.youtube.com/watch?v=YJxl138WWGk



Figure 31.4 Pseudorabies virus (PRV) infection in piglets. Central nervous signs include ataxia and convulsions, observed after intranasal infection of 4- to 6-week-old piglet with virulent PRV (left) and in one naturally infected suckling piglet (right). *Source:* The Friedrich Loeffler Institute (left). Courtesy of Dr. W. Thiel, Detmold Germany (right).

Clinical Presentation – Pigs

✓Signs in gilts and sows are primarily respiratory and fever.

Consequent **reproductive failure** depends on the phase of gestation and includes

-- First trimester: embryonic death or resorption of fetuses and then return to estrus.

-- Second and third trimesters: mummified fetuses, abortion, stillbirth or weak pigs.

Stillbirth, mummification, embryonic death, and infertility (SMEDI)

Unusual Ocular Presentation

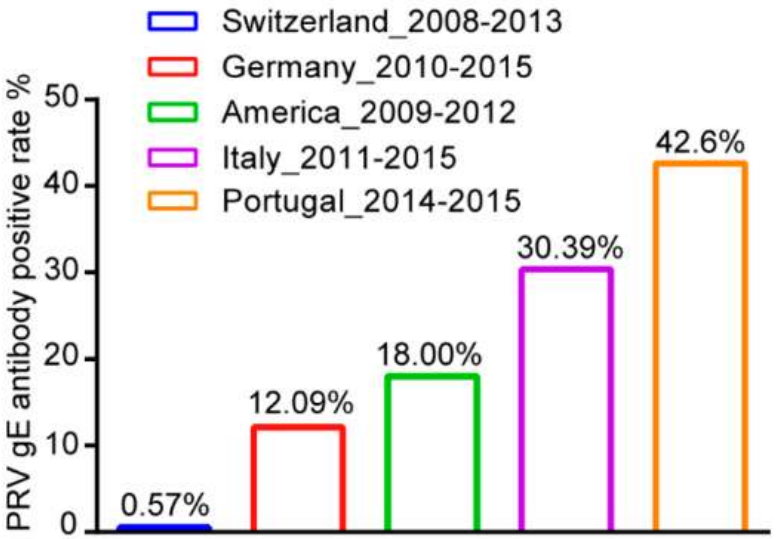


Blindness, corneal opacity, and swelling of the ocular area is observed in a few pigs in each of several litters.

https://www.pig333.com/articles/unusual-ocular-manifestation-of-pseudorabies-in-suckling-piglet_18258/

Clinical Presentation – Wild Boars

- PRV infection in wild boars is usually **asymptomatic**, indicating that circulating PRV variants are **highly adapted** to the host population.
- In Spain and Germany, spontaneous PR in juvenile wild boar have been reported.
- In 2020, a seroprevalence of > 70% was detected in Italian wild boars population.



Liu et al., 2022

Clinical Presentation – Others

- **Per-acute** clinical course, usually die within 48h.
- Mainly **neurological symptoms**; often only extreme **pruritus** and subsequent self-mutilation are observed.
- Dead mice, rats, dogs, or cats on farms are **telltale signs for the presence of PRV** prior to the onset of clinical signs in infected pigs.



Courtesy: Institute of Veterinary Pathology, Justus Liebig University, Giessen

Gross Lesions

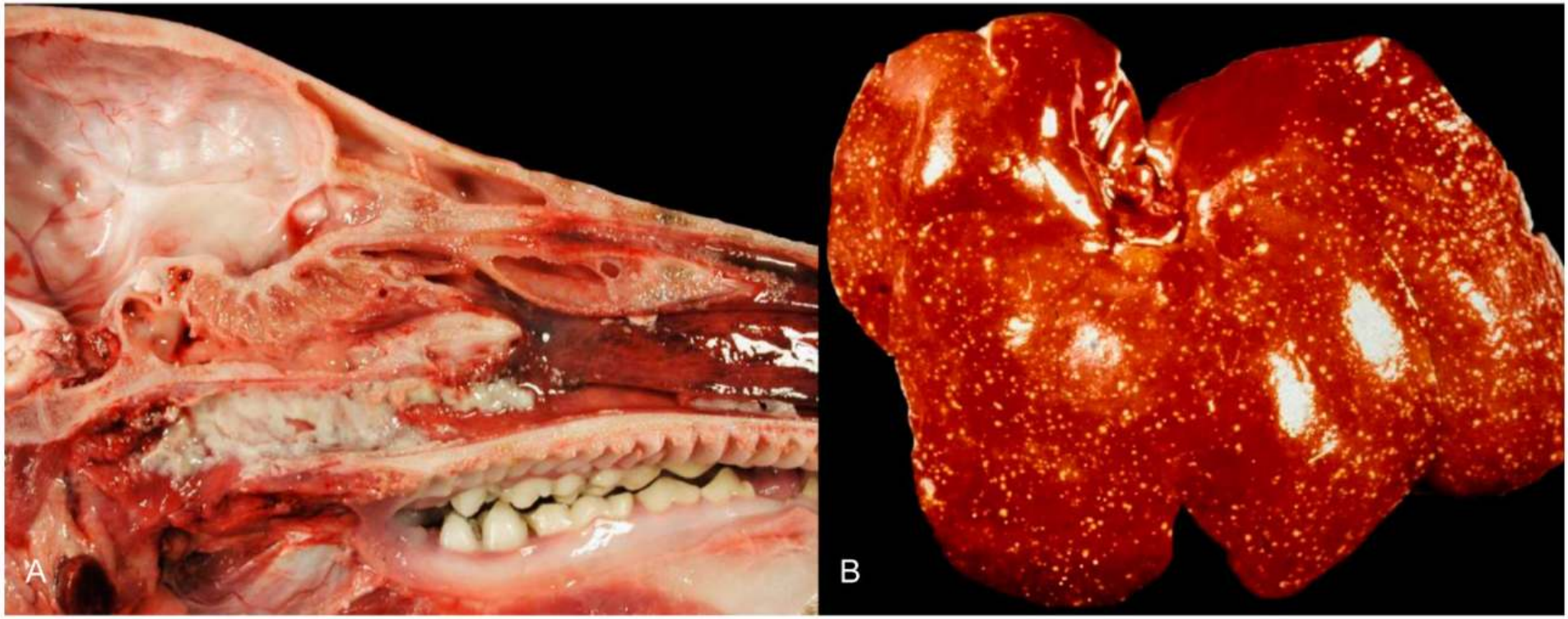
No pathognomonic gross lesion.

- Multiple small foci (1–3mm in diameter) of acute **hemorrhagic necrosis** may be seen in the liver, spleen, lung, intestines, and adrenals (characteristic of alphaherpesvirus infection).
- Exudative keratoconjunctivitis, serous to fibrinonecrotic rhinitis, laryngitis, tracheitis, and necrotizing tonsillitis may be present.
- CNS lesions are inconspicuous except for the congestion of the meninges.

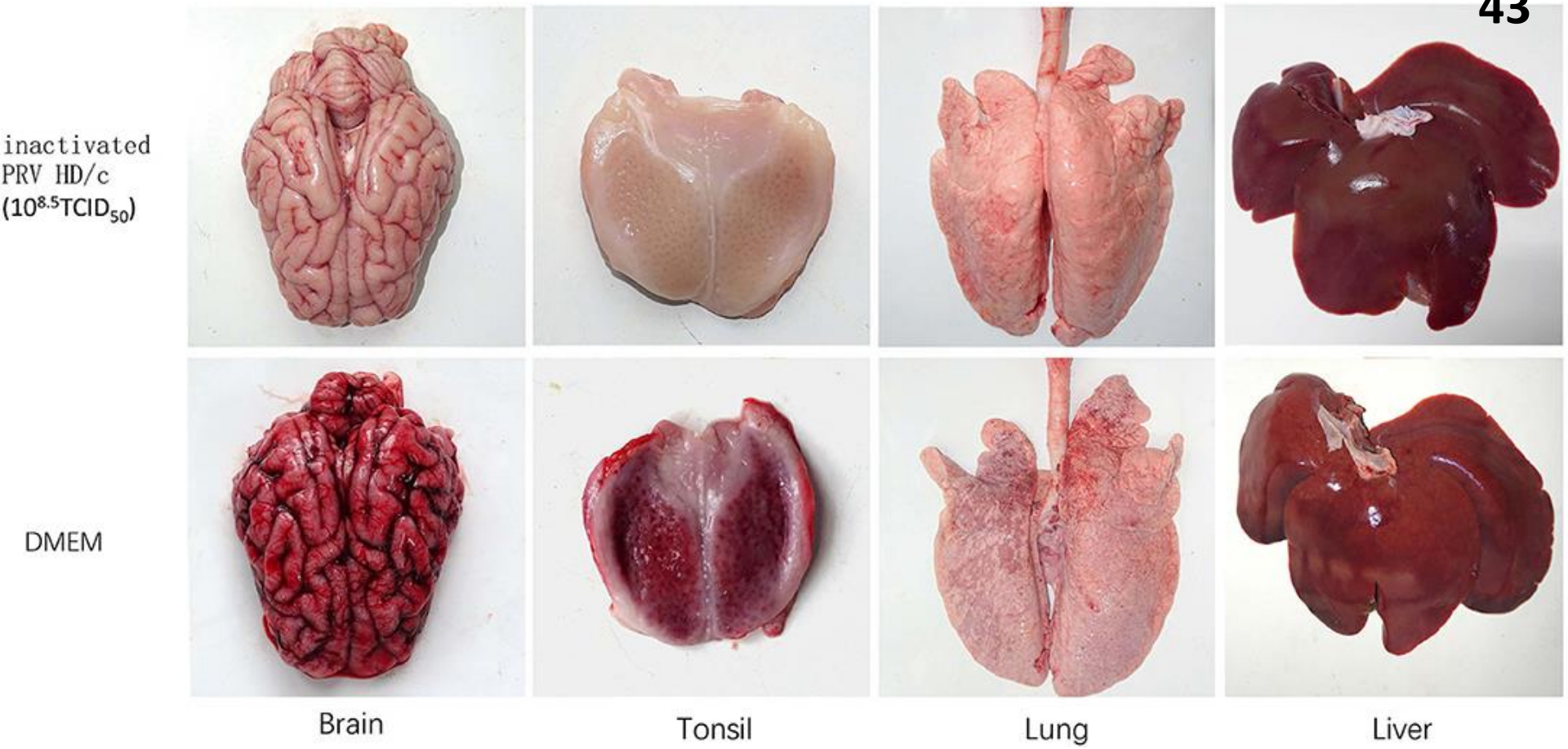
Gross Lesions

- In sows, necrotizing placentitis and endometritis with thickened, edematous wall of the uterus are observed after abortion.
- SMEDI; gross ± microscopic foci of necrosis may be present in aborted fetuses.
- Edema of the scrotal region

Coagulative necrosis of the palatine tonsil (left) and liver (right).



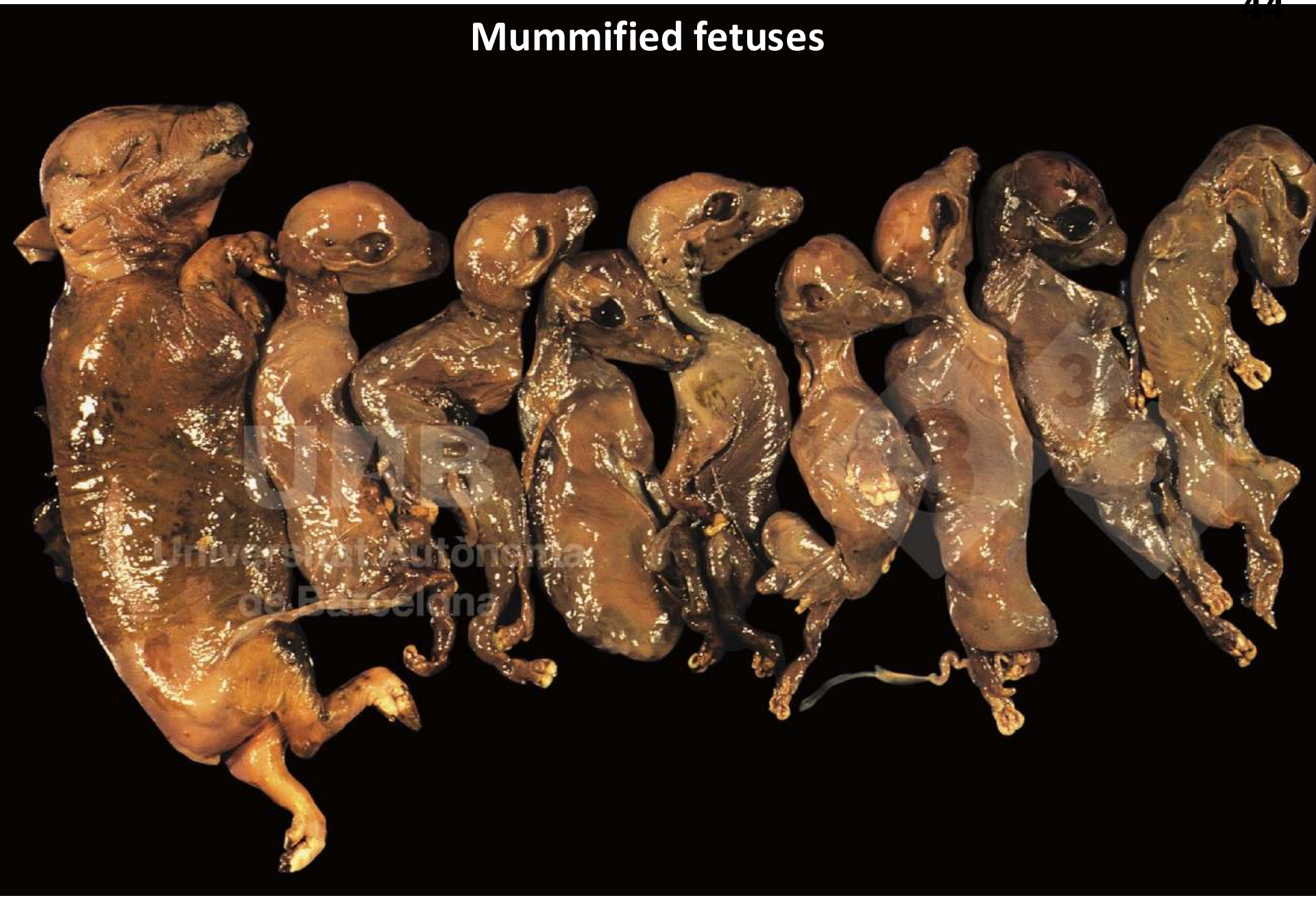
Courtesy: Institute of Veterinary Pathology, Justus Liebig University, Giessen).



Pathological changes of inactivated piglets challenged with the PRV strain DX. Groups of pigs (n = 5 per group) were vaccinated with inactivated PRV HD/c ($10^{8.5}TCID_{50}$) or DMEM, and challenged with the PRV strain DX ($10^{5.4}TCID_{50}$) at 21 dpi. The brain, tonsil, lung, and liver of inactivated piglets were collected and subjected to pathological examinations. Serious cerebral and tonsil hyperemia, many pinpoint hemorrhages in the lungs, and multiple small focal areas of necrosis in and liver were observed.

Jin et al., 2022

Mummified fetuses



https://www.pig333.com/pathology-atlas/mummified-fetuses_56

Microscopic Lesions

Microscopic lesions in pigs reflect the neuroinvasive and epitheliotropic properties of PRV.

Viral intranuclear **inclusion bodies** are much more common in lesions **outside the nervous system.**

- CNS: nonsuppurative meningoencephalomyelitis
- PNS: ganglioneuritis of the trigeminal and paravertebral ganglia
- Epithelial lesions consist of areas of coagulative or lytic, partially hemorrhagic, **necrosis** with the presence of the amphophilic intranuclear inclusions.

Microscopic Lesions

- Female reproductive tract: multifocal to diffuse lymphohistiocytic endometritis and vaginitis and necrotic placentitis with coagulative necrosis of chorionic fossae develop. Intranuclear inclusion bodies are present in degenerate trophoblasts associated with necrotic lesions
- Male reproductive tract : degeneration of seminiferous tubules and necrotic foci in the tunica albuginea of the testicles may be observed. Spermatozoal abnormalities occur.

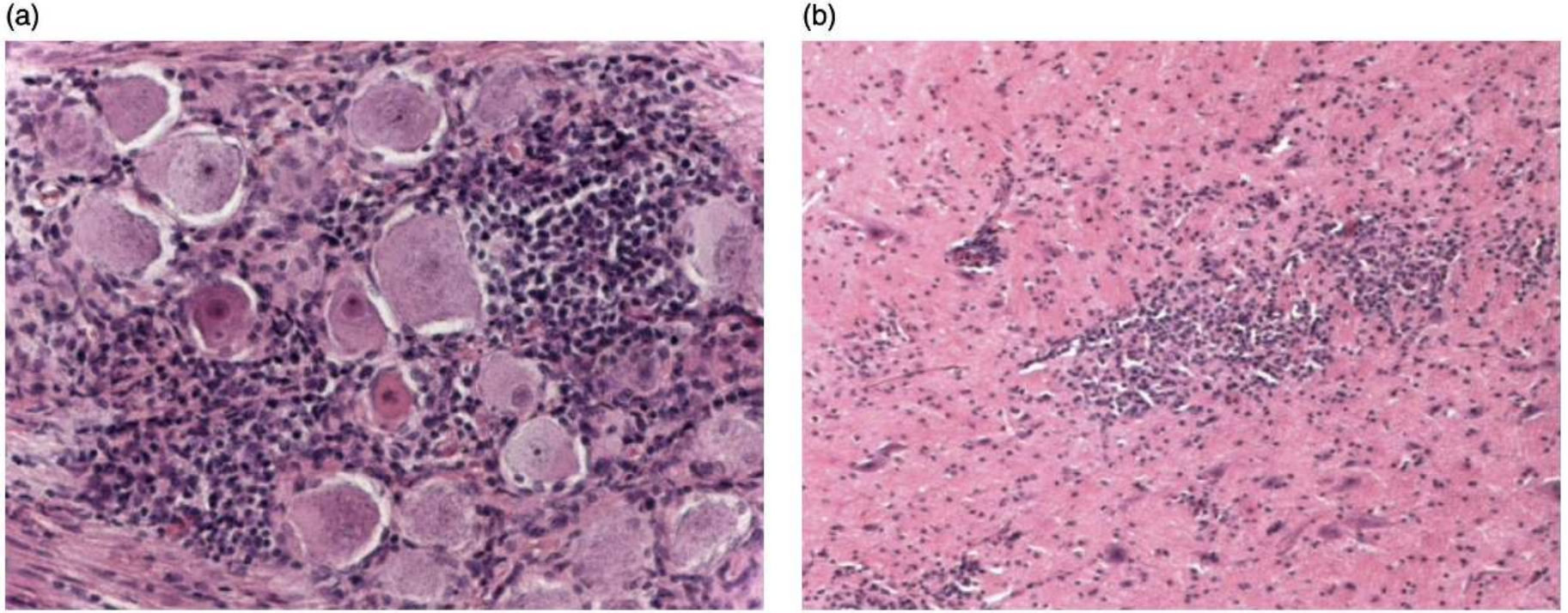


Figure 35.6 (a) Diffuse lymphoplasmacytic ganglioneuritis in the trigeminal ganglion with degeneration and necrosis of perikarya and (b) multifocal-to-coalescing nonsuppurative encephalitis in the medulla with neuronal degeneration and perivascular cuffing 8 days after PRV infection under experimental conditions.

Diseases of Swine, 11th Edition

Differential Diagnosis

- Rabies
- Porcine polioencephalomyelitis(Teschovirus and Sapelovirus infection)
- Classical and African swine fever
- Nipah virus infection
- Japanese encephalitis
- Hemagglutinating encephalomyelitis
- Porcine circovirus (PCV2)
- Highly virulent strains of PRRSV
- Swine influenza
- Encephalomyocarditis (EMC)
- Streptococcal meningoencephalitis
- Erysipelas
- Salt or organic poisoning

Differential Diagnosis

In pigs:

- Neurological signs and mortality in young pigs: African swine fever, classical swine fever, teschovirus, streptococcus suis, salt toxicity
- Respiratory signs: PPRS, influenza, PCV2
- Reproductive signs: Parvovirus, PRRS, PCV2

In other species:

- The PRV is rather strictly **neuroinvasive**, and diseases of the CNS, such as rabies, prion (scrapie in sheep & bovine spongiform encephalopathy in cattle), and diseases causing persistent itching need to be excluded.

Laboratory Diagnosis

Rapid detection of infection is essential for effective PRV control.

Virus detection (Secretions, excretions, and tissues)

Trigeminal ganglia and tonsils for latent infection

- ✓Identification of (infectious) virions: EM, histo, virus isolation
- ✓Identification of antigen: impression smears and cryosections of tissues
- ✓Identification of nucleic acids: PCR, qPCR

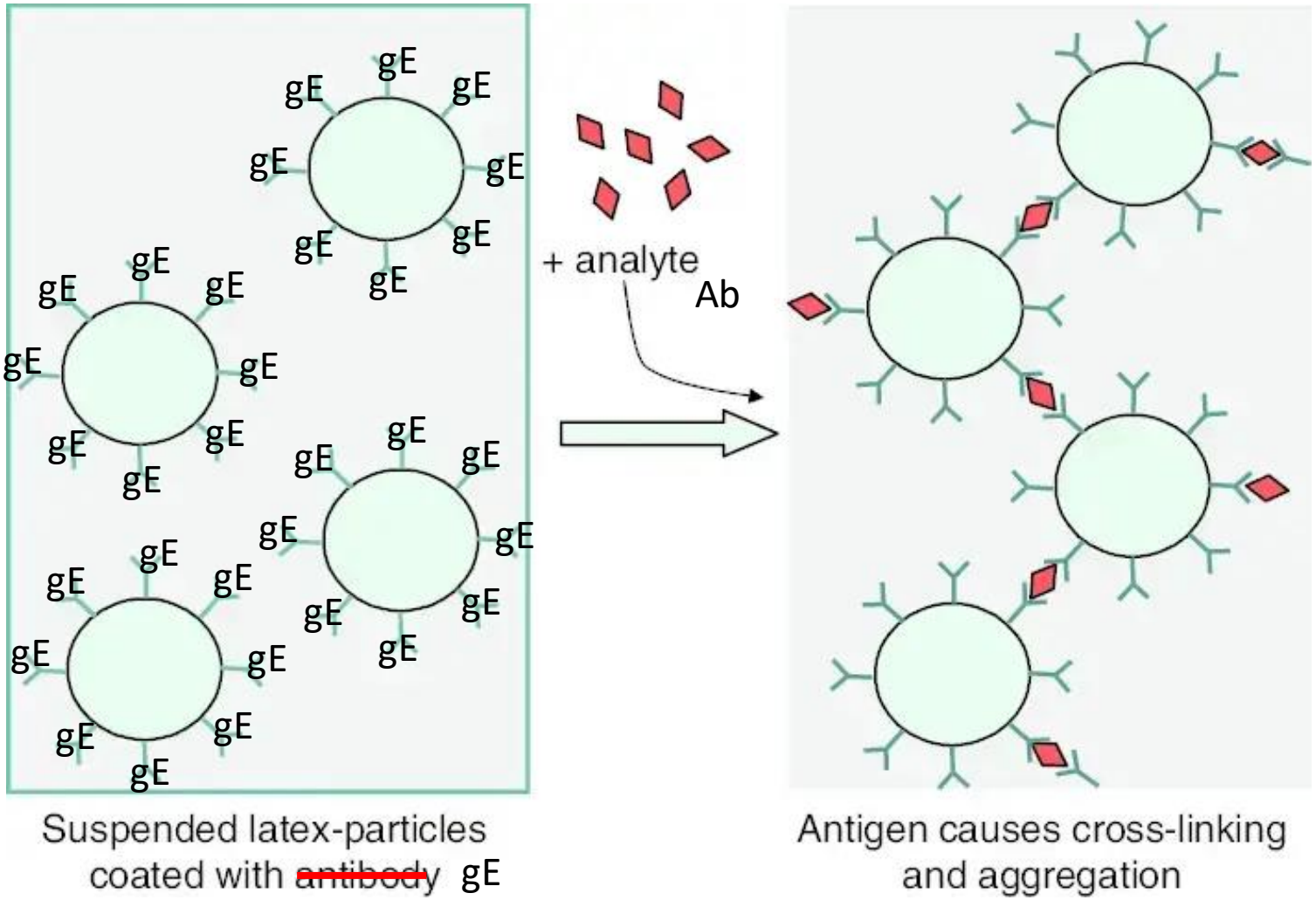
-- Primers target conserved region, such as gB, gC, gD, or gE.

Laboratory Diagnosis

Serological test:

- ✓Virus neutralization, ELISAs, latex agglutination (LAT)
- ✓VN are highly reliable but cannot differentiate between antibodies resulting from natural infection or vaccination.
- ✓ELISAs able to detect serum antibodies against **gE** (or gC, gG) allowed for achieving **DIVA** purpose and became a key part of PRV eradication programs.

Latex agglutination (LAT)



Modified from https://biologynotesonline.com/latex-agglutination-test-procedure-principle-inhibition-limitation-uses/

Prevention & Control

- There is **no specific treatment** for PR.
- **All-in all-out practices with disinfection** between groups help prevent PR infection.
- Eradication can be achieved through **DIVA** vaccination, good management practices, and elimination of animals that carry the disease.

Vaccination

- Does **not prevent primary infection** or development of latency.
- Protects pigs from clinical signs and decreases virus shedding
- Maternally derived immunity prevents the transmission of PRV in newborn piglets and is able to protect against clinical disease after infection by limiting virus replication in the CNS.

Maternally derived antibodies inhibit the ability of piglets to respond to vaccination.

- Attenuated, inactivated, genetically-modified (gE-) vaccines
- **DIVA vaccine + DIVA test → eradication of PR**

DIVA

Whole package:

DIVA vaccine + DIVA test

- Selectable marker (Ag)

  -- In natural infection:

  Good antigenicity and longevity

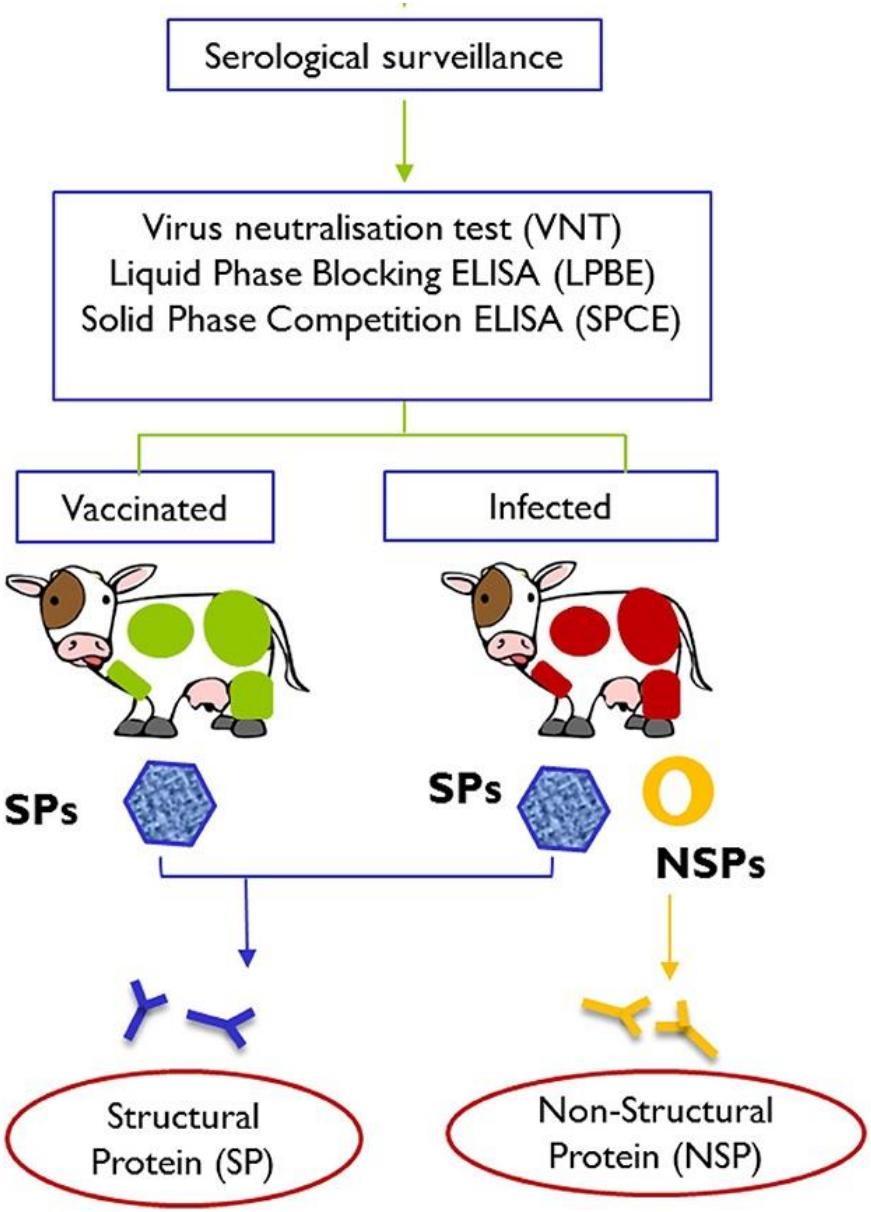
  -- Upon vaccination

  Absence of immunoreactivity

- ex. 3A3B NSP for FMD DIVA vaccine

  NS3 for CSF DIVA vaccine

  gE for PR DIVA vaccine



Wong et al., 2016

Mutations in PrV vaccine strain Bartha

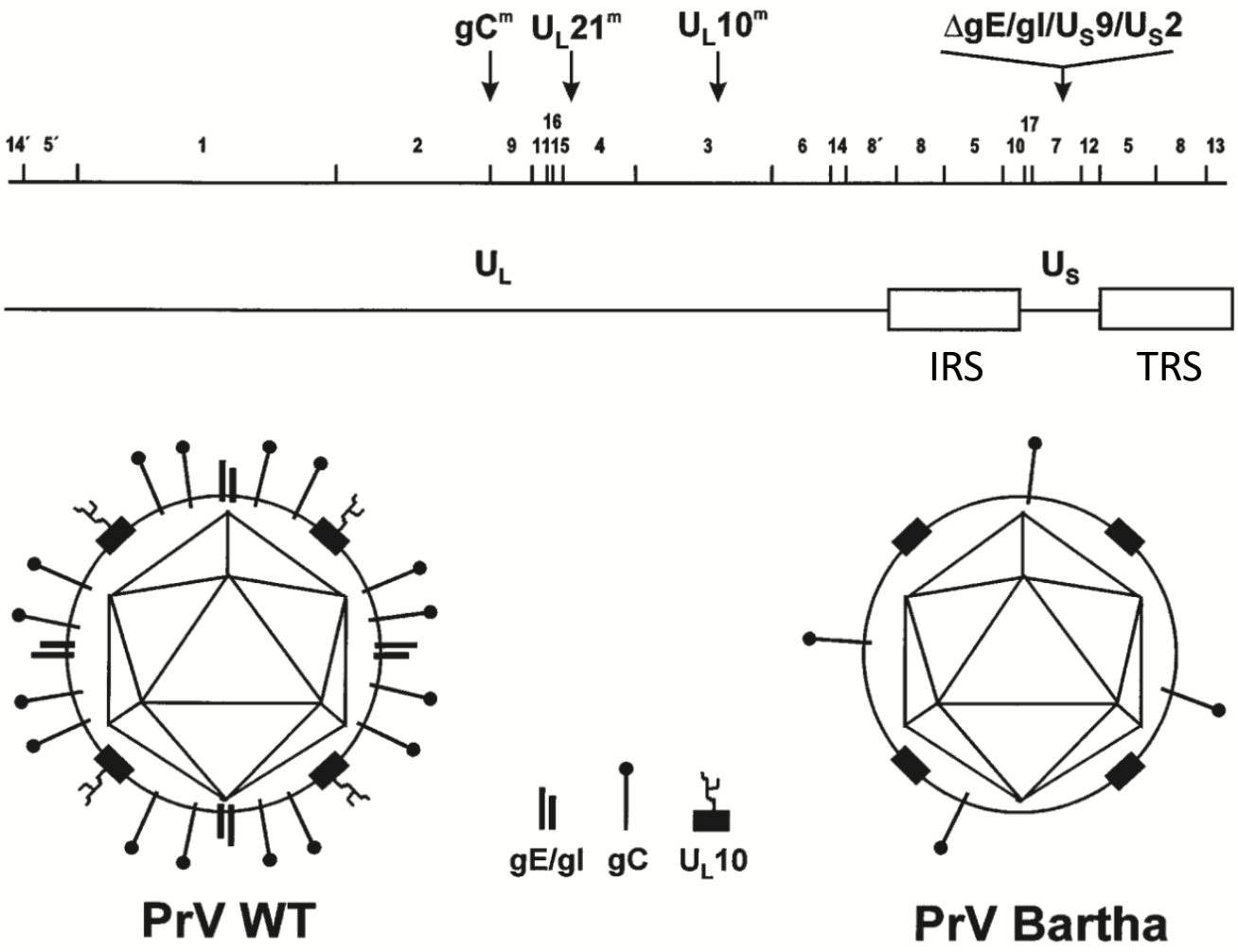
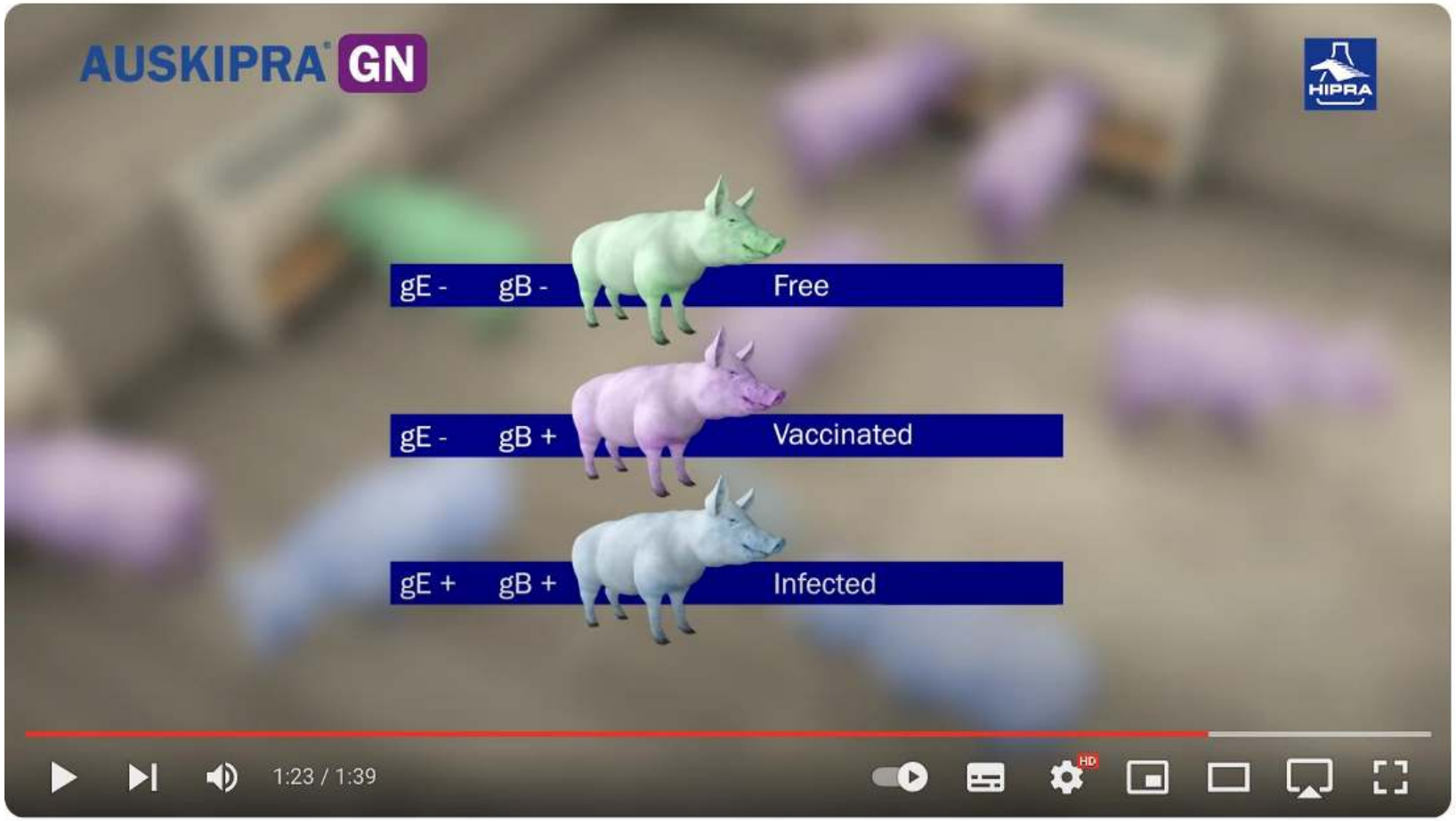


Figure 4. Mutations present in the genome of the PrV strain Bartha. The deletion in the U_S region encompassing the gI, gE, Us9 and Us2 genes, and the mutations in the gC, UL10, and UL21 genes are indicated. The consequences of these mutations on the envelope structure of PrV Bartha are depicted schematically below in comparison with PrV wildtype (WT).

Modified from Mettenleiter, 2020

Gene-Deleted Vaccines	Vaccine Strains	Progenitor Strains	Deleted Gene	Technology Used	Authorization	References
Single gene-deleted vaccine	Omnivac	BUK	TK gene	Natural losses	Licensed	[27]
	2.4N3A	NIA-3 (field strain)	gE gene	HR	Licensed	[166]
	PRV(LA-A^{B})	AH02LA (field strain)	gE gene	BCA	Not available	[167]
	HN1201ΔgE (inactivated)	HN1201 (field strain)	gE gene	HR	Licensed	[154]
	rPRVTJ-delgE	TJ (field strain)	gE gene	HR	Not available	[11]
Double gene-deleted vaccine	Omnimark	Omnivac (BUK)	TK and gIII genes	Natural losses	Licensed	[145]
	Begonia	2.4N3A (NIA-3)	TK and gE genes	Natural losses	Licensed	[168]
	NIA3-783	2.4N3A (NIA-3)	TK and gE genes	HR	Licensed	[146]
	Tolvi	field strain	TK and gpX genes	HR	Licensed	[169]
	D1200/D560	NIA-3	gD and gI genes	HR	Not available	[148]
	AD-YS400	Yangsan (field strain)	TK and gE genes	HR	Not available	[170]
	JS-2012-ΔgE/gI	JS-2012 (field strain)	gE and gI genes	HR	Not available	[171]
	gE-TK-PRV	TNL (field strain)	TK and gE genes	HR	Not available	[172]
	vZJ01ΔgE/gI (inactivated)	ZJ01 (field strain)	gE and gI genes	BCA	Not available	[157]
	PRV (PRV$^{\Delta TK\&gE\text{-}AH02}$)	AH02LA (field strain)	TK and gE genes	HR	Not available	[173]
Triple gene-deleted vaccine	6C2	Field strain	TK, gE and gI genes	HR	Not available	[174]
	SA215	Fa (classical strain)	gE, gI and TK genes	HR	Licensed	[175]
	rSMXΔgI/gEΔTK	Field strain	TK, gE and gI genes	HR	Not available	[176]
	rPRVTJ-delgE/gI/TK-	rPRVTJ-delgE (TJ strain)	TK, gE and gI genes	HR	Not available	[29]
	vPRV HN1201	HN1201 (field strain)	TK, gE and gI genes	HR	Not available	[158]
	gE^{-}/gI^{-}/TK^{-} PRV	HeN1 (field strain)	TK, gE and gI genes	CRISPR/Cas9	Not available	[177]
	rPRV NY-gE^{-}/gI^{-}/TK^{-}	NY (field strain)	TK, gE and gI genes	HR and CRISPR/Cas9	Not available	[155]
		201715 (field strain)	gE, gC and TK genes	CRISPR/Cas9	Not available	[178]
	rPRV/XJ5-gE^{-}/gI^{-}/TK^{-}	XJ5 (field strain)	gE, gI and TK genes	HR	Not available	[159]
	rGXΔTK/gE/gI	GX (field strain)	TK, gE and gI genes		Not available	[165]
Four gene-deleted vaccine	PrV (376)	PrV (376)	gD, gG, gI and gE genes		Not available	[147]
	——	C (field strain)	gI, gE, Us9 and Us2 genes	Natura losses	Licensed	[179]
	PRV GDFS-delgI/gE/US9/US2	GDFS (field strain)	gI, gE, Us9 and Us2 genes	CRISPR/Cas9	Not available	[161]
	rZJ01-ΔgI/gE/TK/UL13	ZJ01	gI, gE, TK and UL13 genes	CRISPR/Cas9	Not available	[29]
Five gene-deleted vaccine	PRV rHN1201$^{TK-/gE-/gI-/11k-/28k-}$	HN1201 (field strain))	TK, gI, gE, 11k and 28k genes	BCA	Not available	[164]

Zheng et al., 2022



https://www.youtube.com/watch?v=3KbeHhNlKwQ

5.下列何者可採用全場疫苗接種、differentiate infected from vaccinated animal（DIVA）篩選及淘汰策略，達到全場清淨？

(A)豬生殖與呼吸綜合症

(B)第二型豬環狀病毒感染症

(C)豬小病毒感染症

(D)豬假性狂犬病

下列何種流產性疾病最有可能引起壞死性胎盤炎及子宮內膜炎病變？

(A) 豬離乳後多系統消耗症
(B) 豬生殖與呼吸綜合症
(C) 豬假性狂犬病
(D) 豬小病毒感染

下列有關豬假性狂犬病之敘述，何者錯誤？

(A) 又稱Aujeszky's disease
(B) 豬為保毒動物
(C) 病原為疱疹病毒
(D) 成豬感染常呈現呼吸症狀且死亡率高

下列那一組織不是假性狂犬病病毒潛伏之主要感染部位？

(A) 三叉神經節
(B) 下視丘
(C) 薦神經節
(D) 扁桃腺

下列何種動物對假性狂犬病病毒感受性最高？

(A) 豬
(B) 牛
(C) 羊
(D) 犬

下列何種疾病，已廣泛使用基因缺損之疫苗來有效預防？

(A) 豬流行性下痢
(B) 第二型豬環狀病毒感染症
(C) 豬生殖與呼吸綜合症
(D) 豬假性狂犬病

下列何種疾病目前已有區分受感染或疫苗免疫動物（differentiate infected from vaccinated animals, DIVA）的技 術？①豬瘟 ②口蹄疫 ③豬生殖與呼吸綜合症 ④豬假性狂犬病

(A) ①②③
(B) ①③④
(C) ②③④
(D) ①②④

流產小豬可見全身臟器多發性壞死，肝細胞內有嗜酸性核內包含體，最可能的病原為：

(A) 豬假性狂犬病病毒
(B) 豬生殖與呼吸綜合症病毒
(C) 豬第二型環狀病毒
(D) 豬小病毒

Porcine Cytomegalovirus

Introduction

- Initially designated as inclusion body rhinitis (IBR)
- The virus **grew slowly** in cell culture and produced cytomegaly with large intranuclear inclusions.
- Infections are ubiquitous and **largely ignored** by herd producers.
- In general, clinical infection is rare but fatal systemic disease could develop in young piglets.
- Infection of pregnant, naïve sows may lead to **reproductive failure**: delivery of dead, mummified, stillborn or weak piglets.

台灣省畜衛試所研報 No.23：103～108（1987）

首次發生於台灣之豬細胞巨大病毒感染症

楊喜吟　呂榮修　黎南榮　李淑慧　鄺懋勁
廖永剛　李　全　林地發

台灣省家畜衛生試驗所

1987年10月間，台灣北部某企業化養豬場一批5週齡小豬350頭陸續發病，在三個星期的病程中死了47頭，死亡率為13.43％。臨床症狀僅見精神沈鬱及呼吸稍急促。剖檢病變以大量胸水（400～500 ml）及心囊液（30～50ml）、肺小葉間中腸顯著水腫及鼻粘膜潮紅為主。組織病理變化則以非化膿性腦炎、鼻粘膜管泡腺上皮細胞腫脹5至10倍及含有特大之不定形嗜鹼性核內包涵體為特徵。鼻粘膜乳劑及其粘液性排出物經遠心及負染色後，於電子顯微鏡下觀察到直徑約110nm的無外膜病毒粒子及250 nm的有外膜病毒粒子。

本病由其特徵性的病理變化及電子顯微鏡檢查結果，診斷為豬細胞巨大病毒感染症（porcine cytomegalovirus infection）。此為本病在台灣發生之首次報告。

Etiology

Porcine cytomegalovirus (PCMV) or suid herpesvirus 2 (SuHV–2)

- Genus *Roseolovirus*, subfamily of *Betaherpesviridae*
- **Enveloped** & characteristics herpesvirus morphology
- Double-stranded sense DNA genome, approximately 130 kb encoding at least 79 ORFs.
- Only **one serotype**

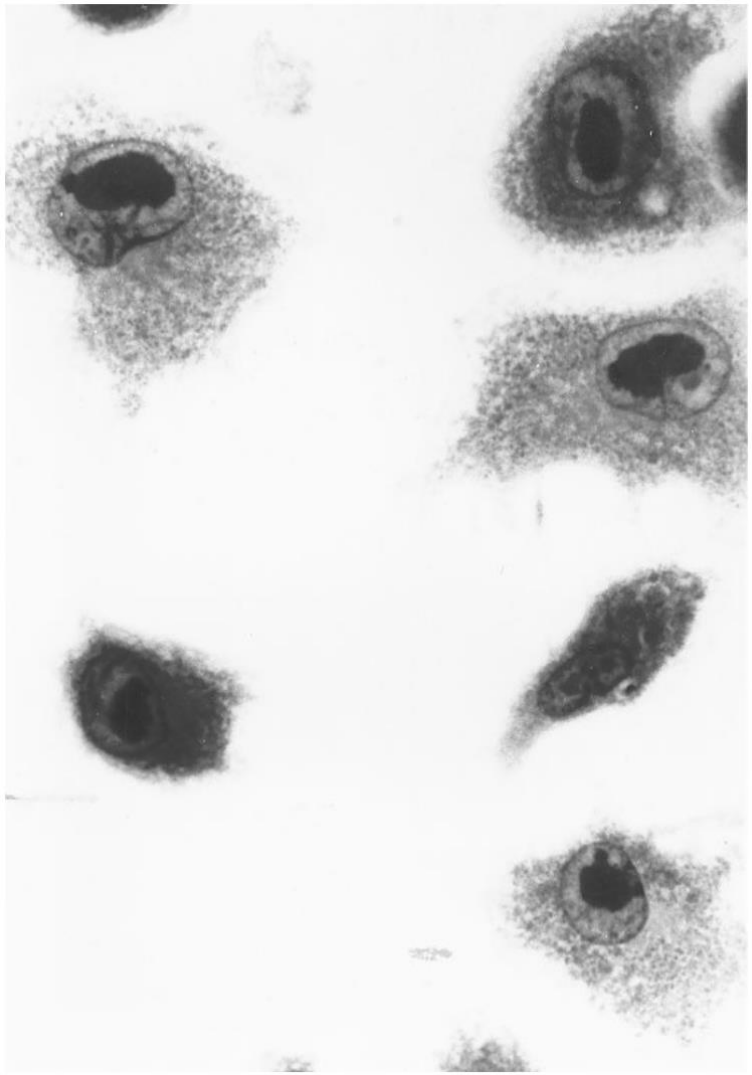


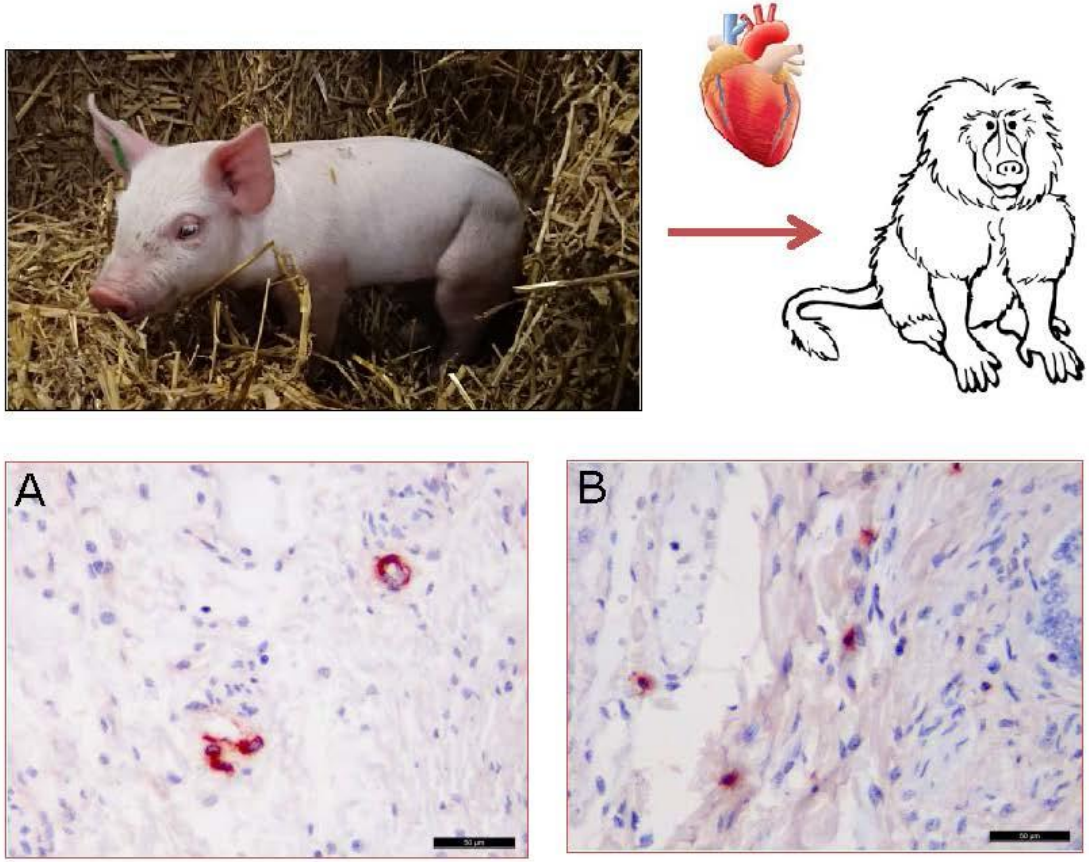
Figure 35.7 Cultured pig macrophages 11 days after inoculation with PCMV. Note the basophilic intranuclear inclusions in the enlarged cells (May–Grünwald–Giemsa; ×720). *Source:* courtesy of R.G. Watt.

Diseases of Swine, 11th Edition

Epidemiology

- Highly prevalent, herd prevalence > 90% worldwide
- Natural infection with PCMV is limited to pigs
- Non-porcine reservoirs and arthropod vectors have not been reported.
- **Potential public concern**: cross-species transmission of PCMV was reported in pig-to-primate **xenotransplantation**.

Distribution of Porcine Cytomegalovirus in Infected Donor Pigs and in Baboon Recipients of Pig Heart Transplantation



Xenotransplantation using pig cells, tissues and organs may be associated with the transmission of porcine viruses, including the porcine cytomegalovirus (PCMV). Transmission of PCMV has been observed in several cases of transplantation of pig organs into non-human primates and this was associated with reduced transplant survival. Here we describe for the first time the distribution of PCMV in different organs of the donor pig and the baboon recipient. Immunohistochemistry using an antiserum that is specific for PCMV detected virus protein-expressing cells in all of the organs of the recipient baboon, most likely representing disseminated pig cells. The graphic shows a multitransgenic pig generated for xenotransplantation, the process of heart transplantation together with PCMV-positive cells in the transplanted pig heart (A) and in the baboon liver (B).

Fiebig et al.,2018

Transmission

- Virus is shed in **nasal and ocular secretions, urine and the aerosol** created by sneezing and coughing.
- **Horizontal**: inhalation or ingestion of virus via oronasal route
- **Vertical**: probably crosses the placenta to infect developing fetuses
- Recovered animals develop latency and can then become carrier or shedders under stressed conditions.
- PCMV establish **latency in porcine monocytes/macrophages** and CD8+ lymphocytes.

Pathogenesis

- PCMV can infect epithelial cells, monocyte/macrophages, and endothelial cells.
- PCMV replicate primarily in **epithelial cells of nasal mucosa and/or the lachrymal or Harderian glands** → cell-associated viremia →
    1. In nursing or growing pigs: nasal mucosal glands, Harderian and lachrymal glands, kidney tubules, and, more rarely, the epididymis and mucous glands of the esophagus.
    2. In fetus or neonate: there is predominantly infection of capillary endothelium and sinusoids of lymphoid tissues, thus resulting in systemic spread of PCMV and generalized lesions

Clinical Presentation

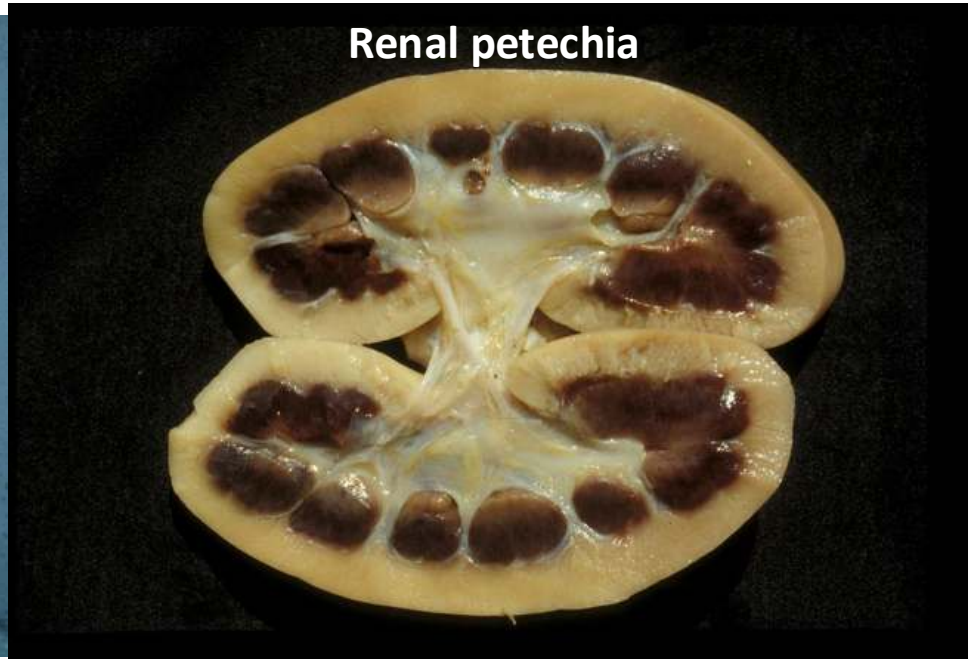
The disease is generally self-limiting and clinical disease usually occurs in 1-5 w pigs with IBR mainly affecting 3-5w pigs.

- Congenitally infection may lead to **reproductive failure** and **weakness and stunting in neonatal piglets** with generalized disease. Edema around the throat and tarsal joints may occur.
- In piglets younger, signs include mucopurulent **rhinitis** accompanied by violent sneezing, respiratory distress, conjunctivitis, shivering, , ± neurological signs, and a variable death loss.
- Infection in older pigs is mainly **subclinical** if uncomplicated.
- Infection with PCMV may exacerbate PRDC.

Lesion

Gross lesion are primarily observed in piglets less than 3 weeks of age with systemic PCMV infection, and may include catarrhal rhinitis, hydrothorax and hydropericardium, pulmonary and subcutaneous edema, and renal petechia.





https://pigdise.nvri.gov.tw/diseshow.aspx?cid=0&docid=26EDD24FE9AD5C1D

Hydrothorax



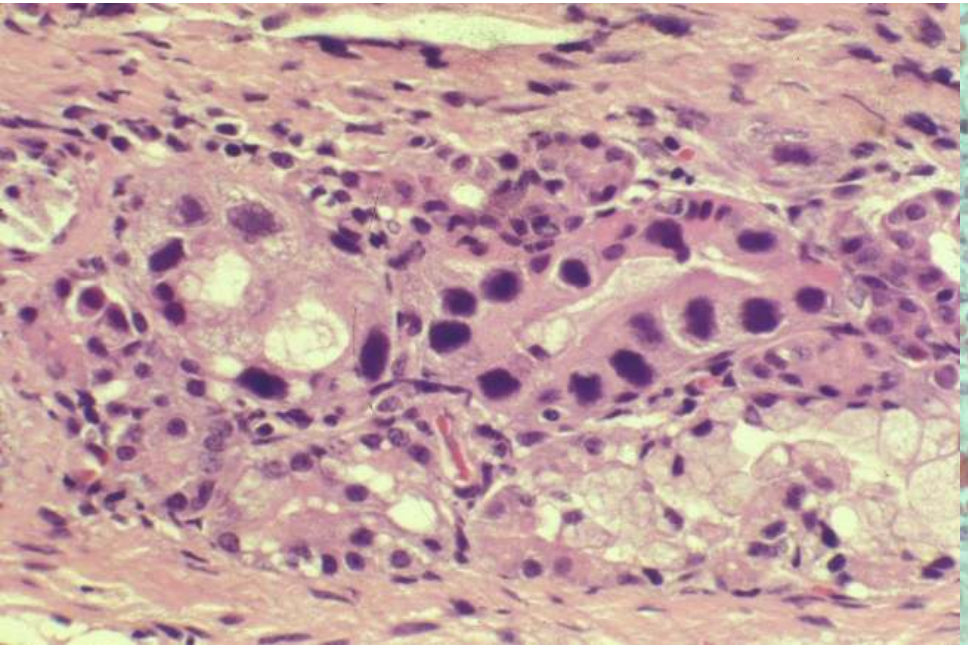
Pulmonary edema

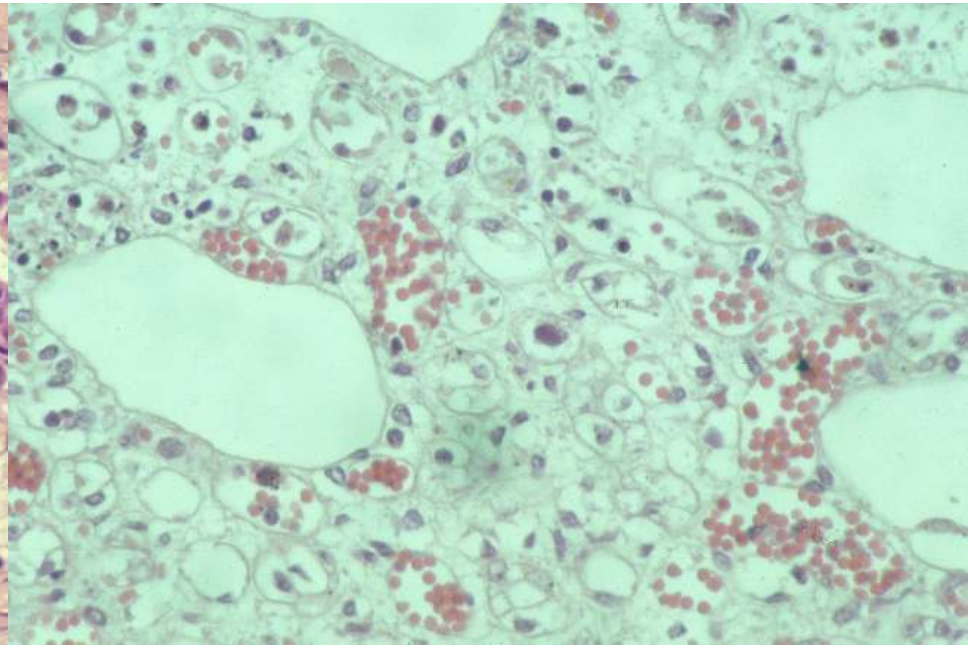


https://pigdise.nvri.gov.tw/diseshow.aspx?cid=0&docid=26EDD24FE9AD5C1D

Lesion

Microscopically, 8–12μm **basophilic intranuclear inclusion bodies**, **cytomegaly**, and **karyomegaly** are seen in the nasal mucous glands, acinar and duct epithelium of Harderian and lachrymal glands, and renal tubular epithelium.





https://pigdise.nvri.gov.tw/diseshow.aspx?cid=0&docid=26EDD24FE9AD5C1D

Diagnosis

The combination of inclusion bodies, cytomegaly, and karyomegaly is pathognomonic

- Identification of (infectious) virion: **Histopath**, virus isolation

  -- Preferred postmortem samples are turbinate mucosa, lungs, pulmonary macrophages obtained by lung lavage, and kidneys.

- Identification of nucleic acids: PCR, qPCR

- If reproductive problems have occurred in a herd, PCMV will need to be differentiated from other reproductive diseases, especially parvoviral infection, pseudorabies (PRV) and porcine reproductive and respiratory syndrome (PRRS).

Prevention & Control

- No vaccine or specific treatment for PCMV is available.
- Introduction of new stock into herds poses a significant risk due to reactivation of latent infections or primary infection of susceptible herds.
- Reduce environmental stressors.

下列有關豬隻感染巨大細胞病毒（cytomegalovirus）之敘述，何者錯誤？

(A)可於Harderian glands複製
(B)可引起肺水腫
(C)可於絲球體之血管內皮發現嗜鹼性核內包涵體
(D)為引起萎縮性鼻炎之主要病原

下列有關豬包含體鼻炎之敘述，何者正確？

(A) 病因為porcine cytomegalovirus
(B) 通常感染成豬
(C) 病毒在上皮細胞大量增殖，形成質內包含體
(D) 本病和細菌混合感染時，造成漿液性滲出物

豬隻感染巨細胞病毒（cytomegalovirus），於鼻黏膜上皮細胞可見何種特徵性病變？

(A) 多發小型嗜酸性質內包涵體
(B) 大型單一嗜酸性質內包涵體
(C) 呈葡萄串狀嗜鹼性質內包涵體
(D) 大型單一嗜鹼性核內包涵體

下列有關豬隻感染細胞巨大性病毒（cytomegalovirus）之敘述，何者正確？

(A) 可於鼻黏膜腺體上皮發現嗜酸性核內包涵體
(B) 主要感染肥育期豬隻
(C) 可見肺臟多發性出血
(D) 可見肺水腫